

Applied Machine Learning with Python & Analytics

(29th March 2025 - 17th August 2025)

in association with
iHUB **Seaborn** and
Ritvij Bharat Private Limited

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Seaborn is a statistical plotting library specifically designed to interact well with Pandas DataFrames.

- Seaborn is built directly off of Matplotlib
- It uses a simpler “one-line” syntax.

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- However, we trade-off customization for ease of use.
- Since its built directly off of Matplotlib, we can actually still make plt method calls to directly affect the resulting seaborn plot.

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- A typical seaborn plot uses one line of code, for example:
 - **`sns.scatterplot(x='salary',y='sales',data=df)`**

Seaborn takes in a pandas DataFrame and then the user provides the corresponding string column names for x and y (depending on the plot type)

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<https://seaborn.pydata.org/>

Common student question:

- *“How do I choose which plot to use?”*

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- It depends on what questions or relationships you are trying to understand.
- At the end of this section, you will have a good intuition of which plots to use.

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Section Topics:

- Scatter Plots
- Distribution Plots
- Categorical Plots
- Comparison Plots
- Seaborn Grids
- Matrix Plots

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Scatter Plots
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Scatter plots show the relationship between two continuous features.

Continuous features are numeric variables that can take any number of values between any two values.

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Continuous Feature Examples

- Age
- Height
- Salary
- Temperature
- Prices

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A **continuous** feature allows for a value to always be between two values.

On the other hand, **categorical** features represent distinct and unique categories:

- Colors
- Shapes
- Names

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Scatter plots line up a set of two continuous features and plots them out as coordinates.

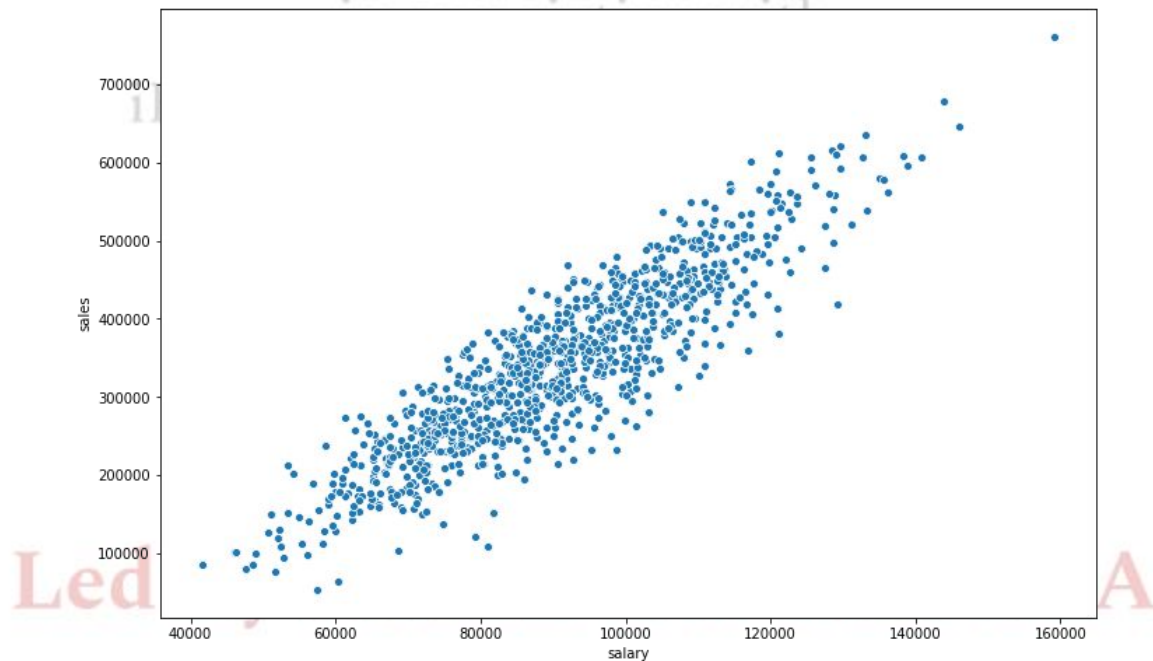
For example, imagine employees with salaries who sell a certain dollar amount of items each year. We could explore the relationship between employee salaries and sales amount.

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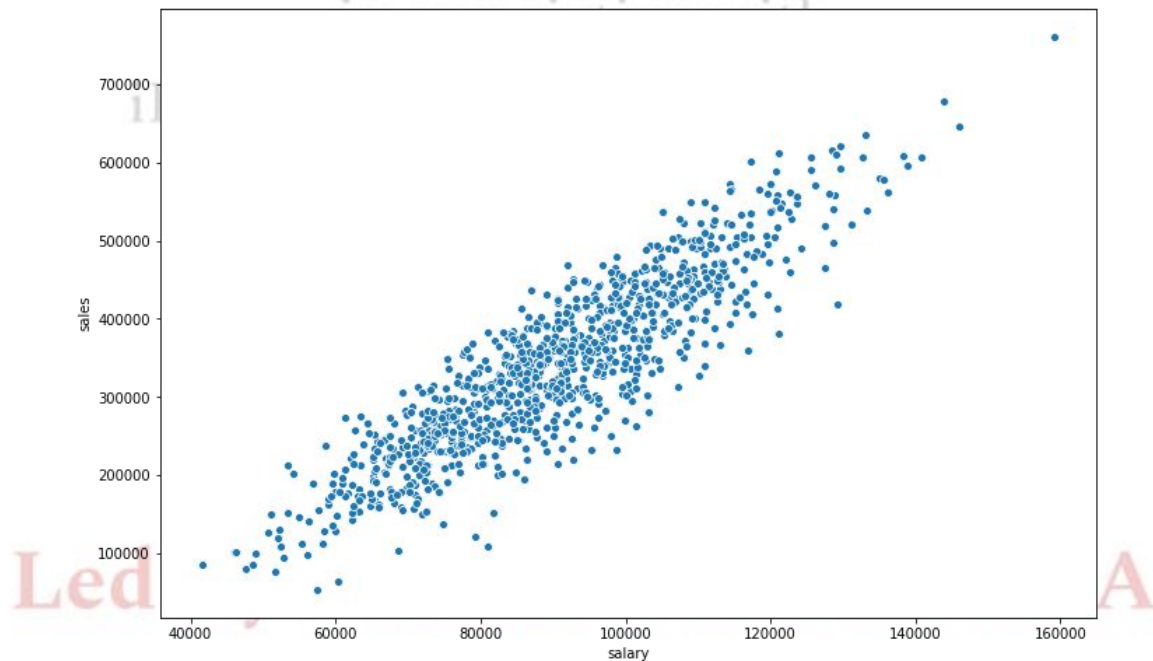
Plot (x,y) coordinate points



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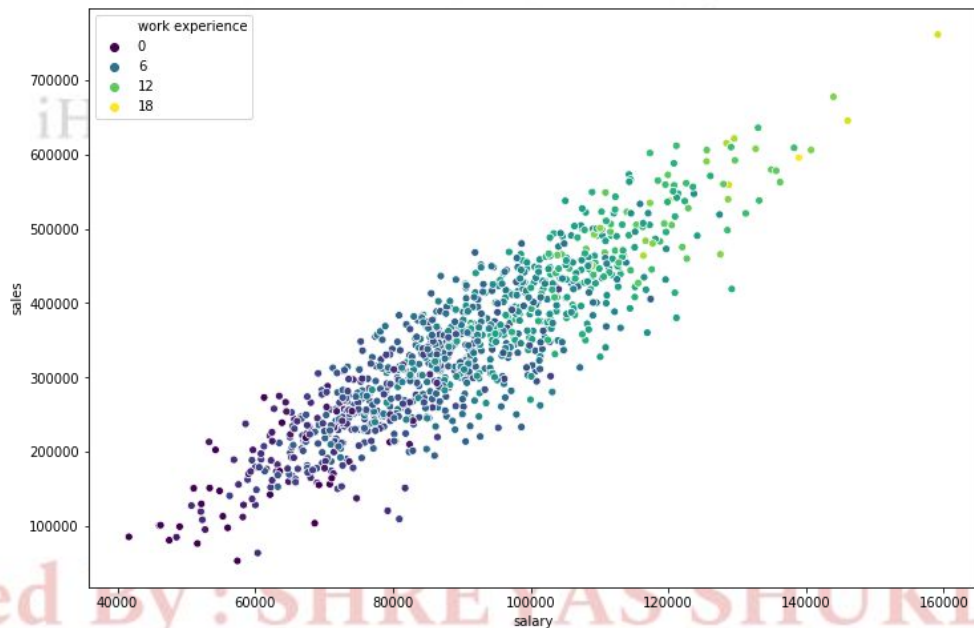
Plot (salary,sales) coordinate points



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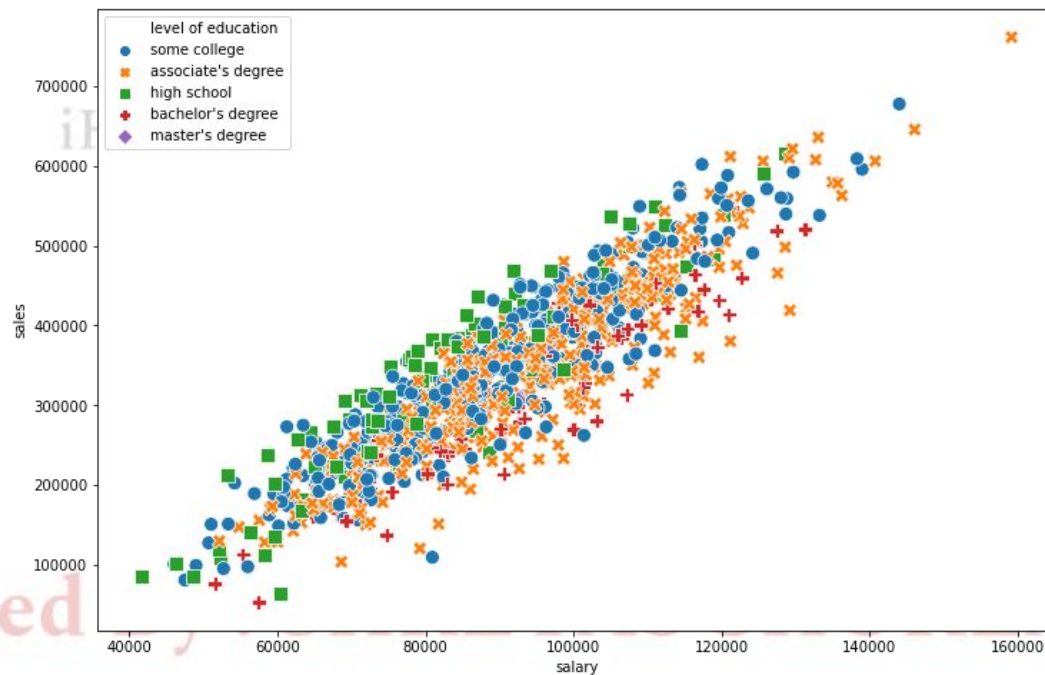
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Seaborn can then add coloring and styling



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Let's explore Scatter Plots with seaborn!

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Distribution Plots

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Distribution plots display a single continuous feature and help visualize properties such as deviation and average values.

3 main distribution plot types:

- Rug Plot
- Histogram
- KDE Plot

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Let's explore the distribution of employee salaries.

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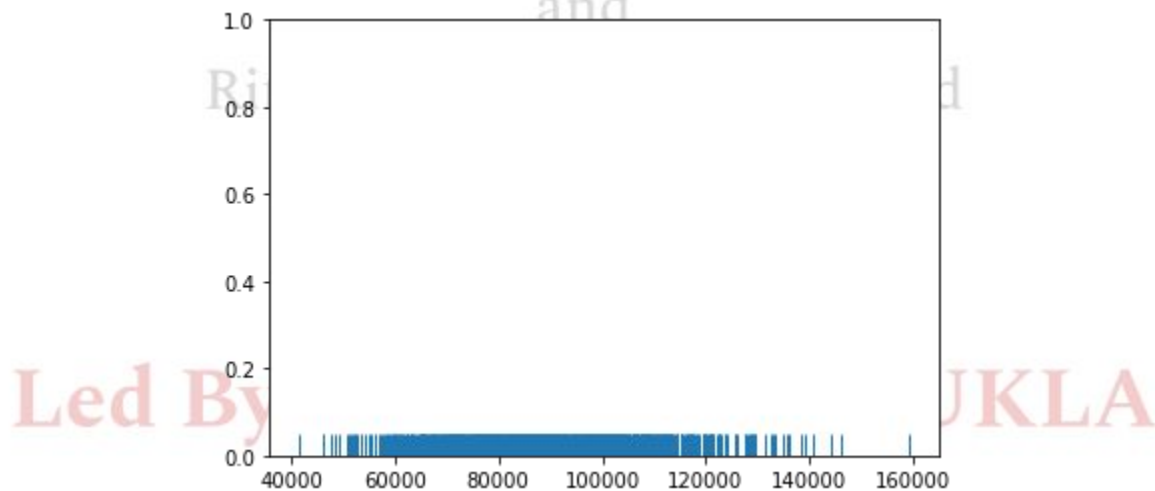
- One way is a rug plot which is the simplest distribution plot and merely adds a dash or tick line for every single value.
- The y-axis does not really have a meaning.

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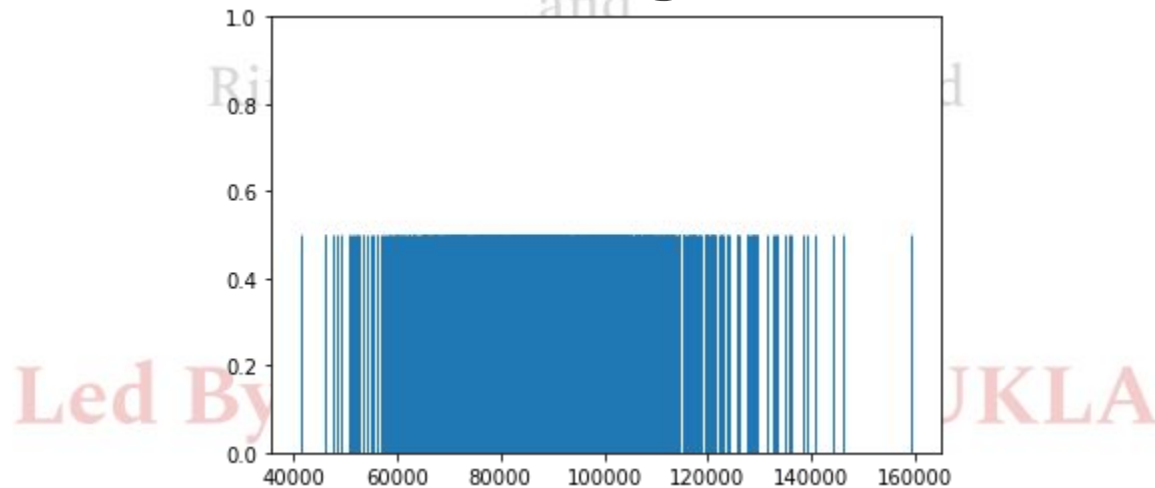
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1. Adds a tick for every salary value
2. Optionally adjust height of ticks
3. Y-axis not interpretable



1. Highest salary near \$160,000
2. Many salaries between \$60k - \$120k

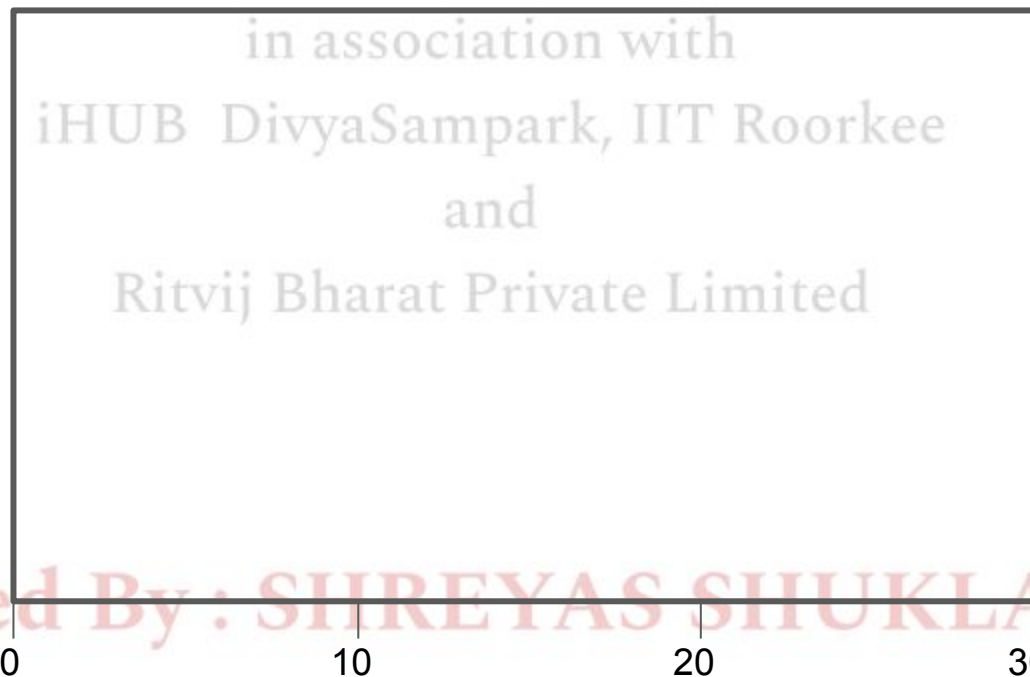
Many ticks could be right on top of each other, we can't tell! If we count how many ticks there are per various x-ranges, we can create a histogram.



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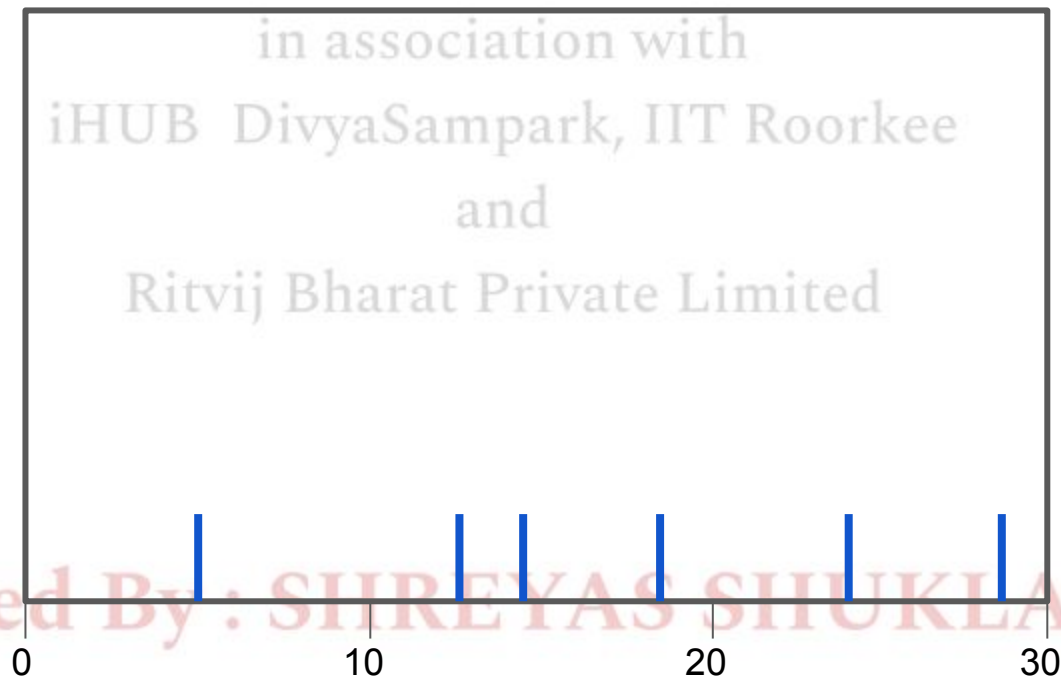
Let's explore a simple example



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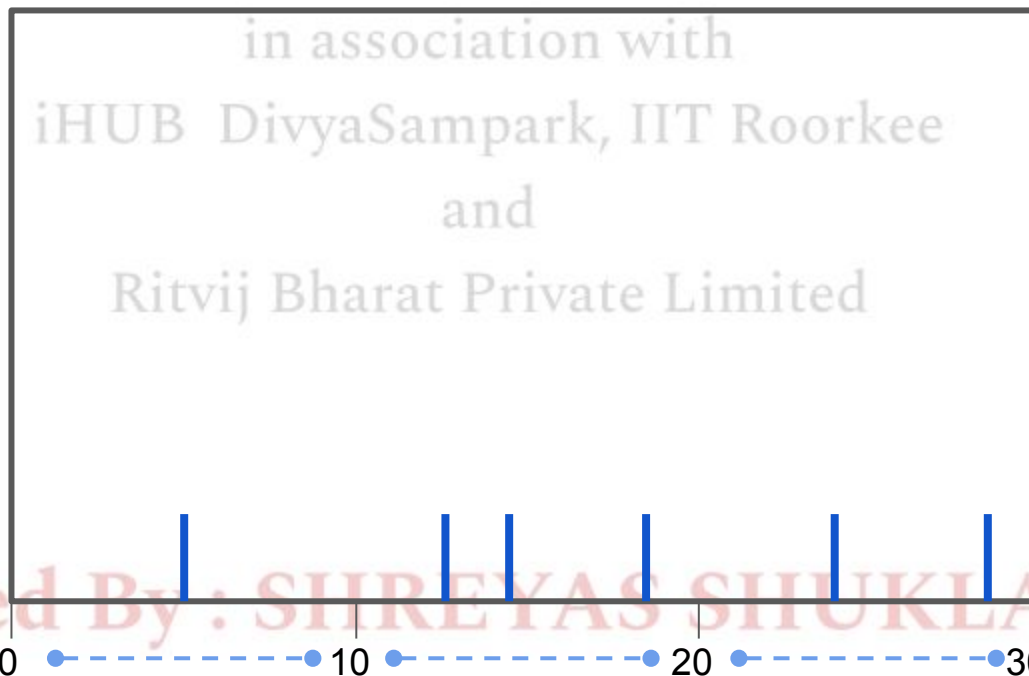
We place the rug plot ticks



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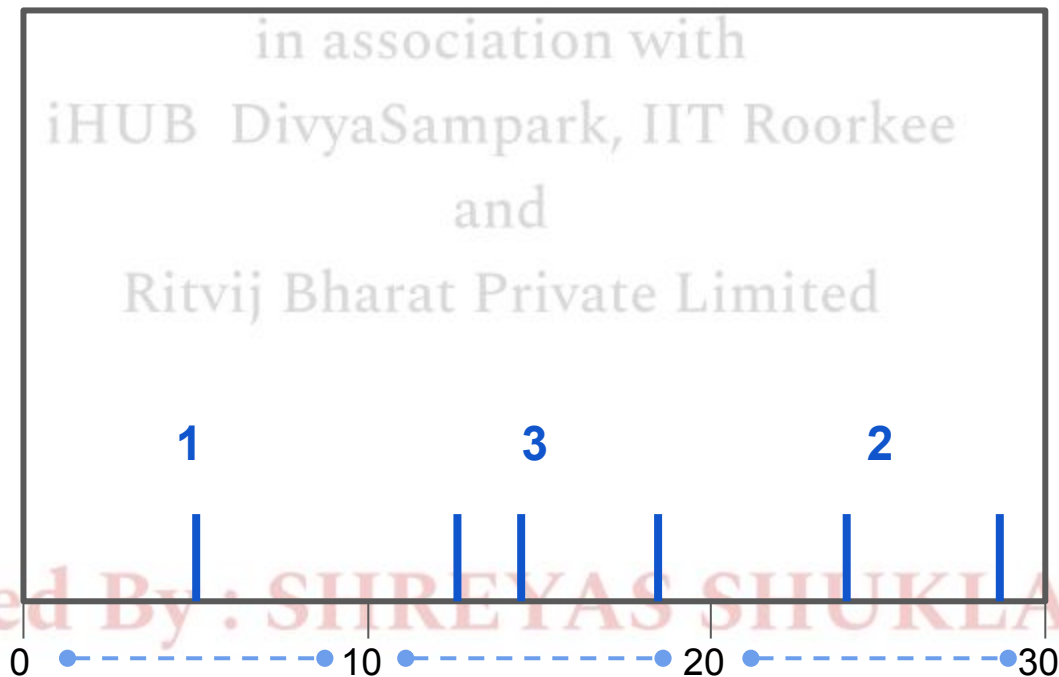
Choose a number of “bins”, we’ll pick 3



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Count ticks per bin

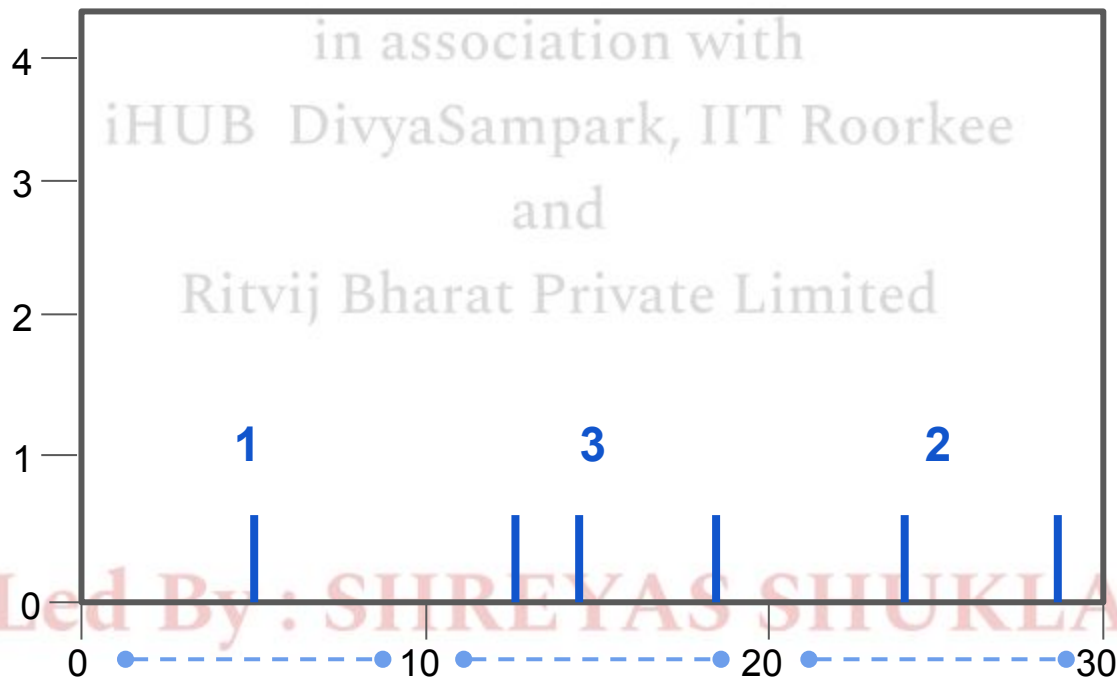


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Create a bar as high as count

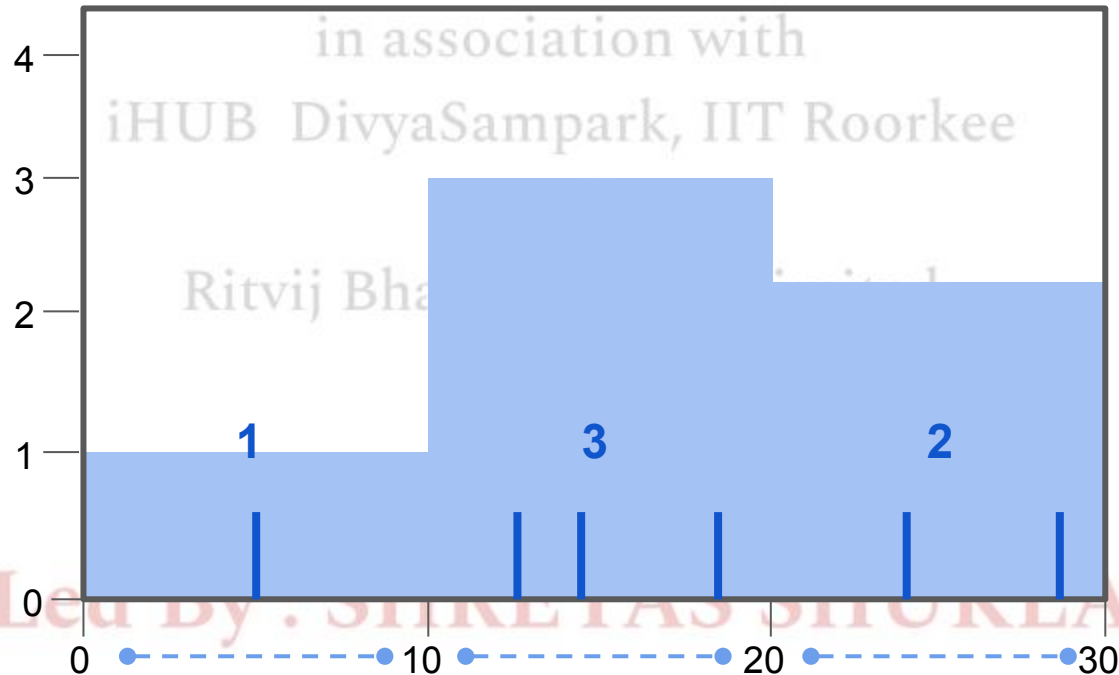


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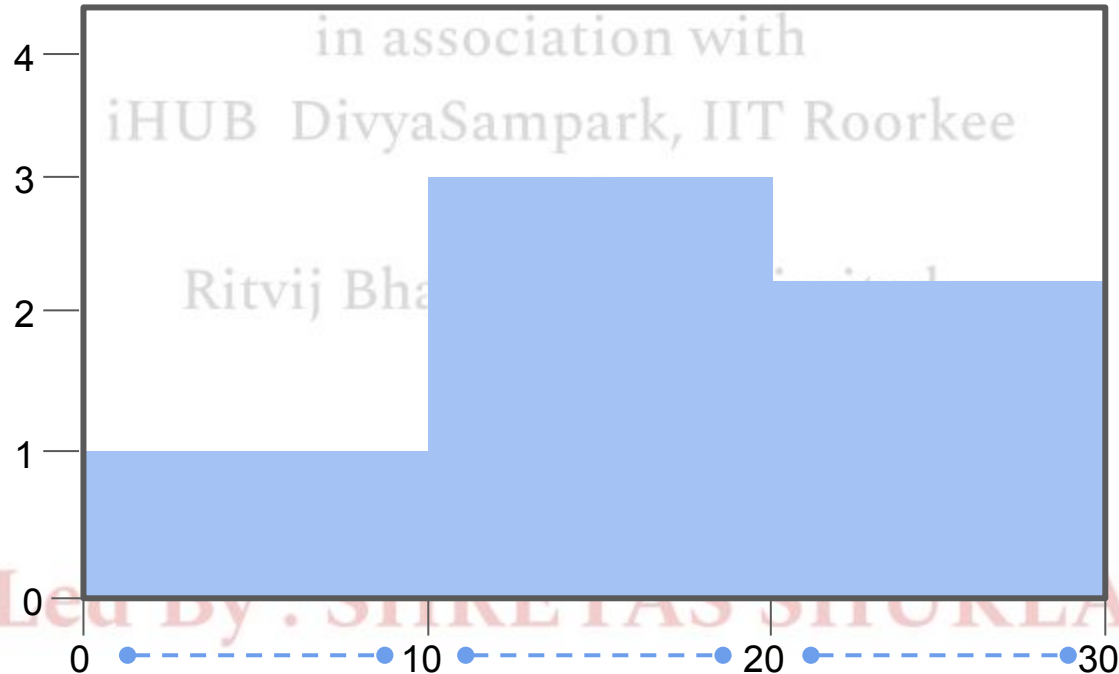
Create a bar as high as count



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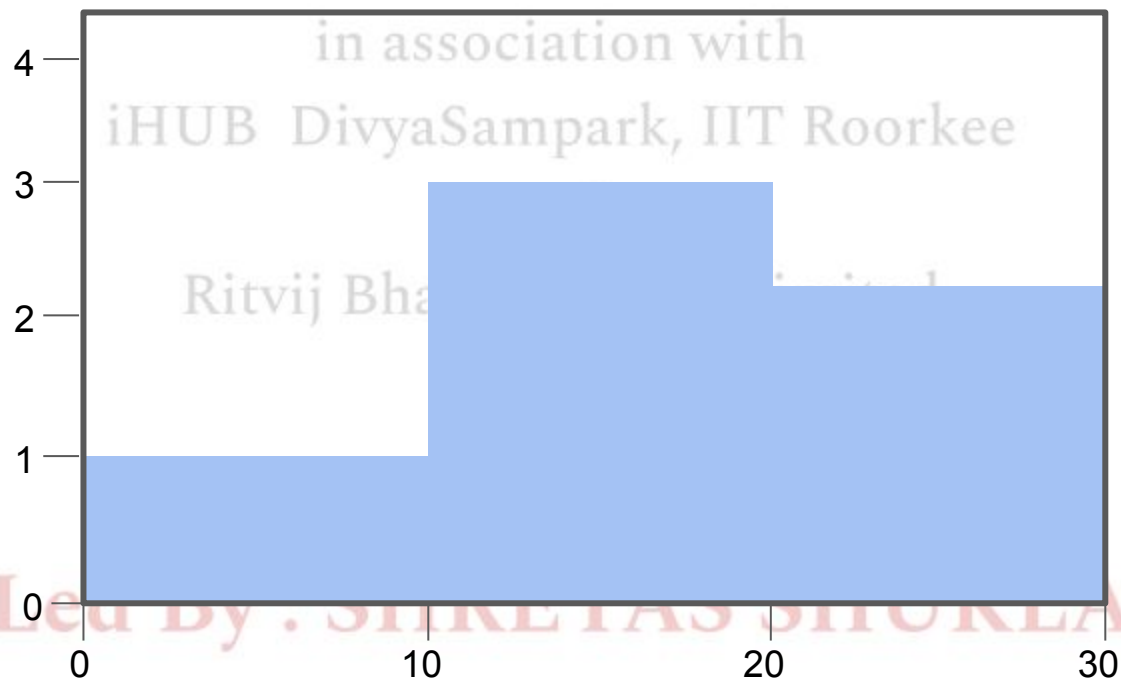
Create a bar as high as count



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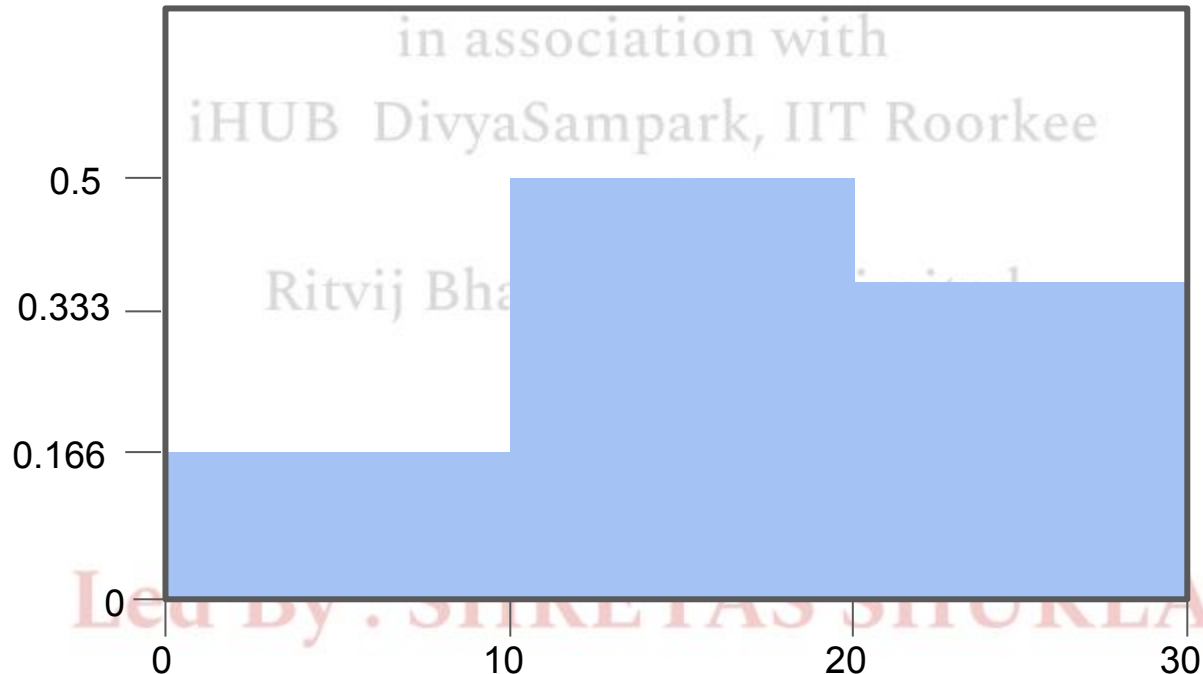
Histogram is complete



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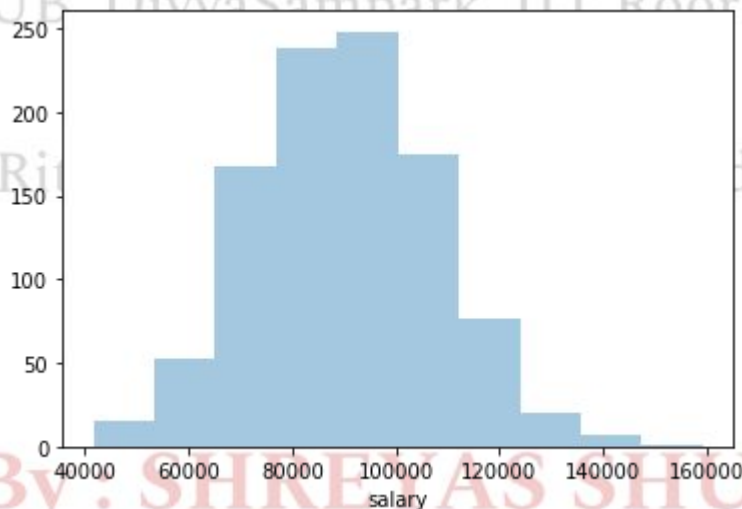
Y-axis can also be normalized as percent



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Changing number of bins shows more detail instead of general trends.

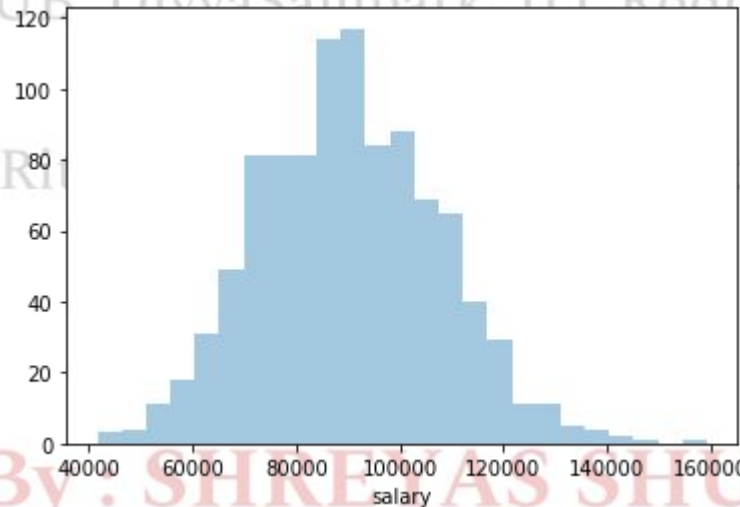


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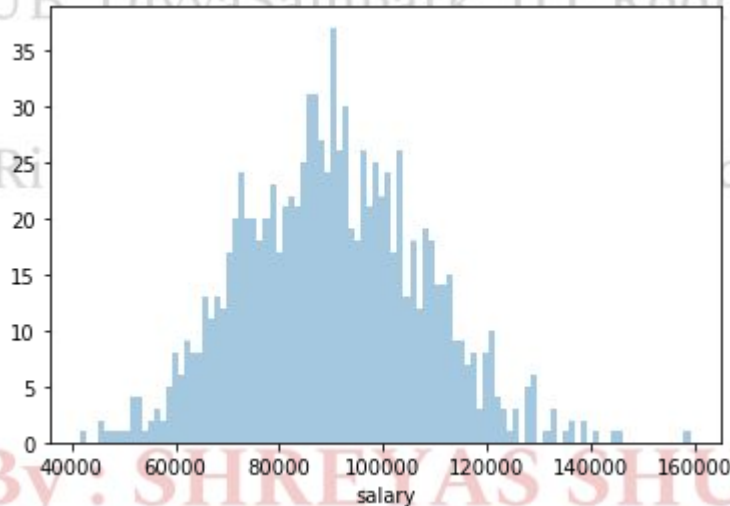


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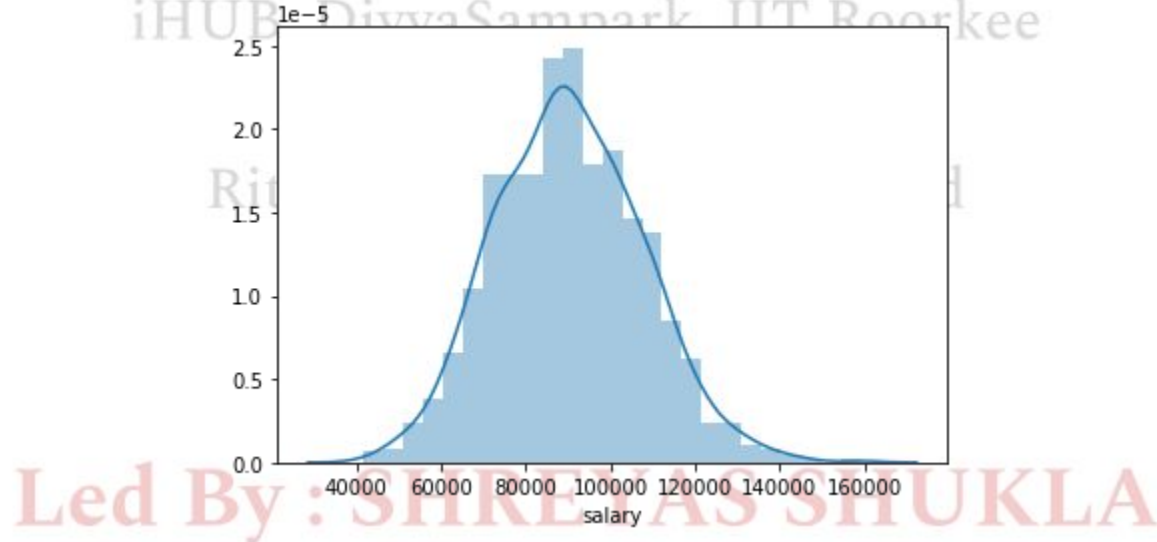


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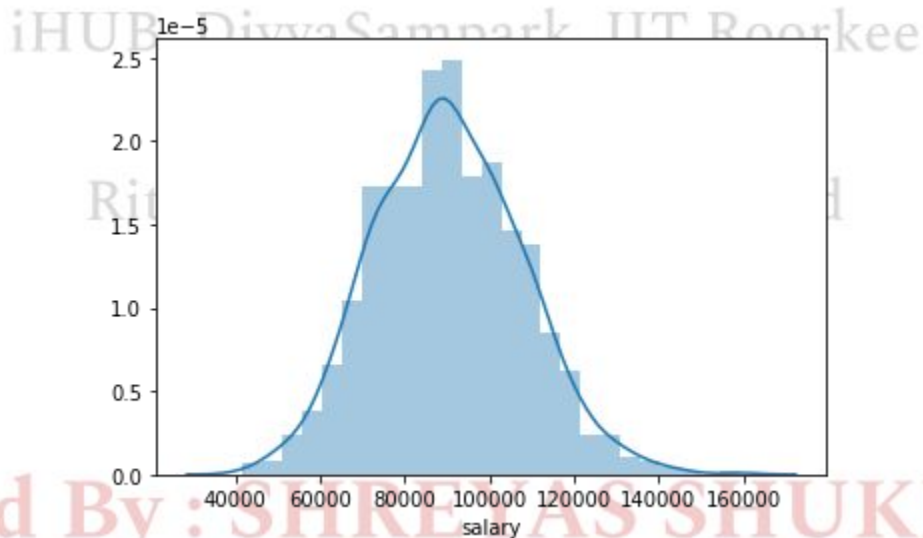
Seaborn also allows us to add on a KDE plot curve on top of a histogram.



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Let's explore what a KDE plot is and how it is constructed.



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Kernel Density Estimation (KDE) is a method of **estimating** a probability density function of a random variable.

In simpler terms, it is a way of estimating a continuous probability curve for a finite data sample.

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KDE plots are best understood by visualizing their
“construction”.

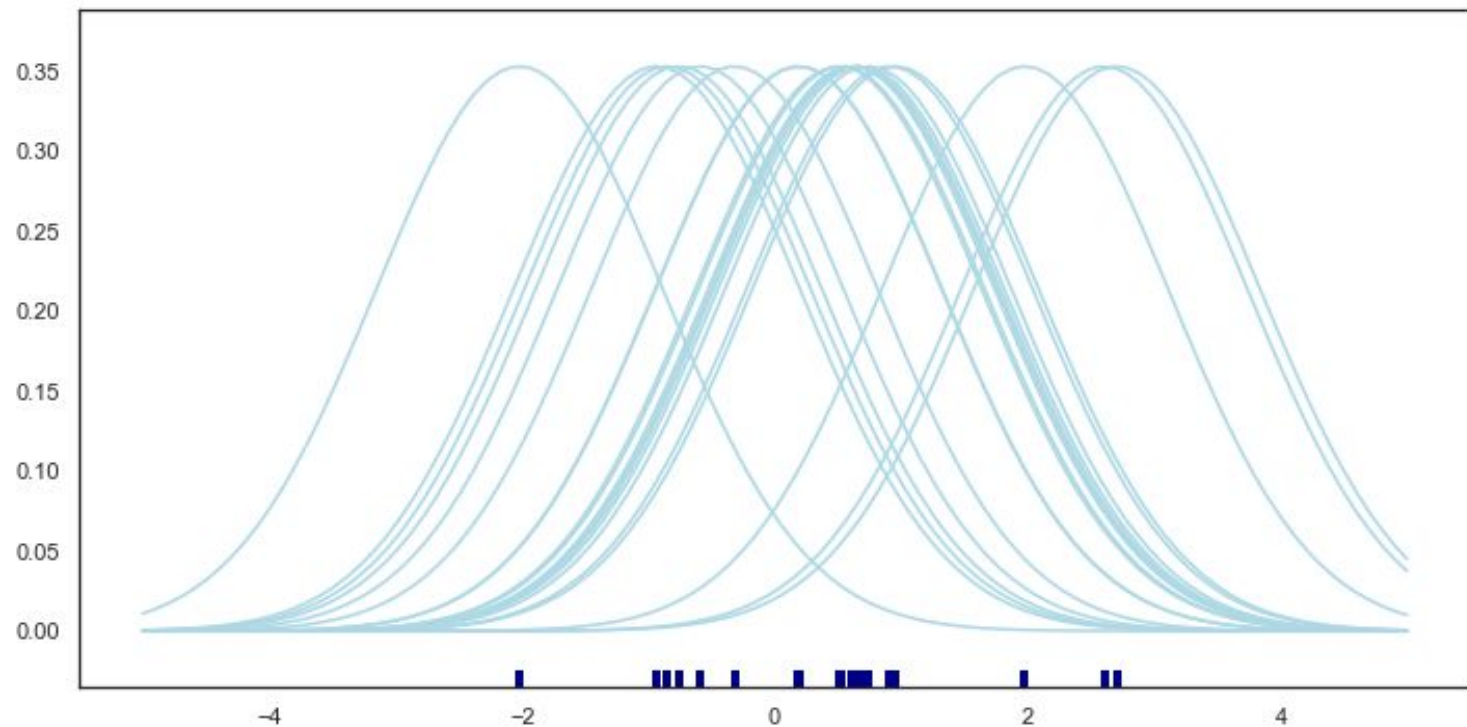
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Let's start with a rug plot....
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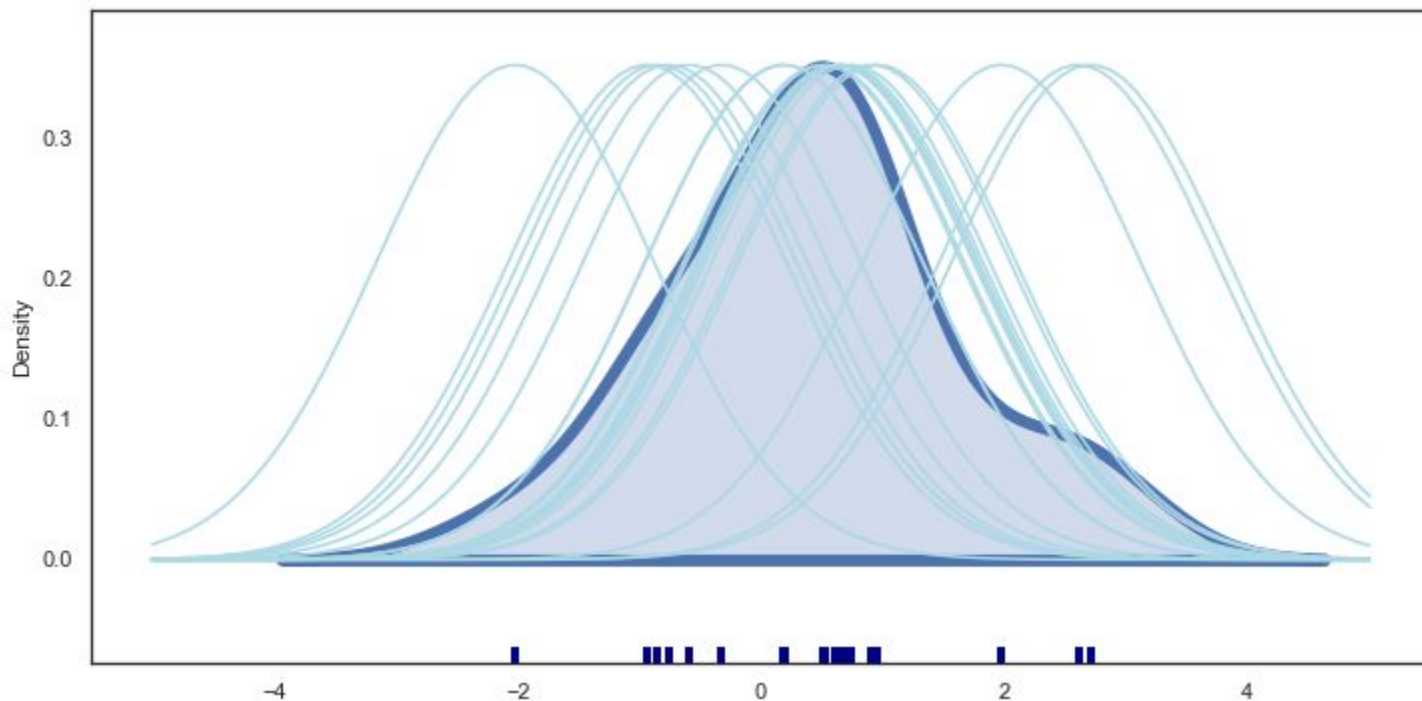
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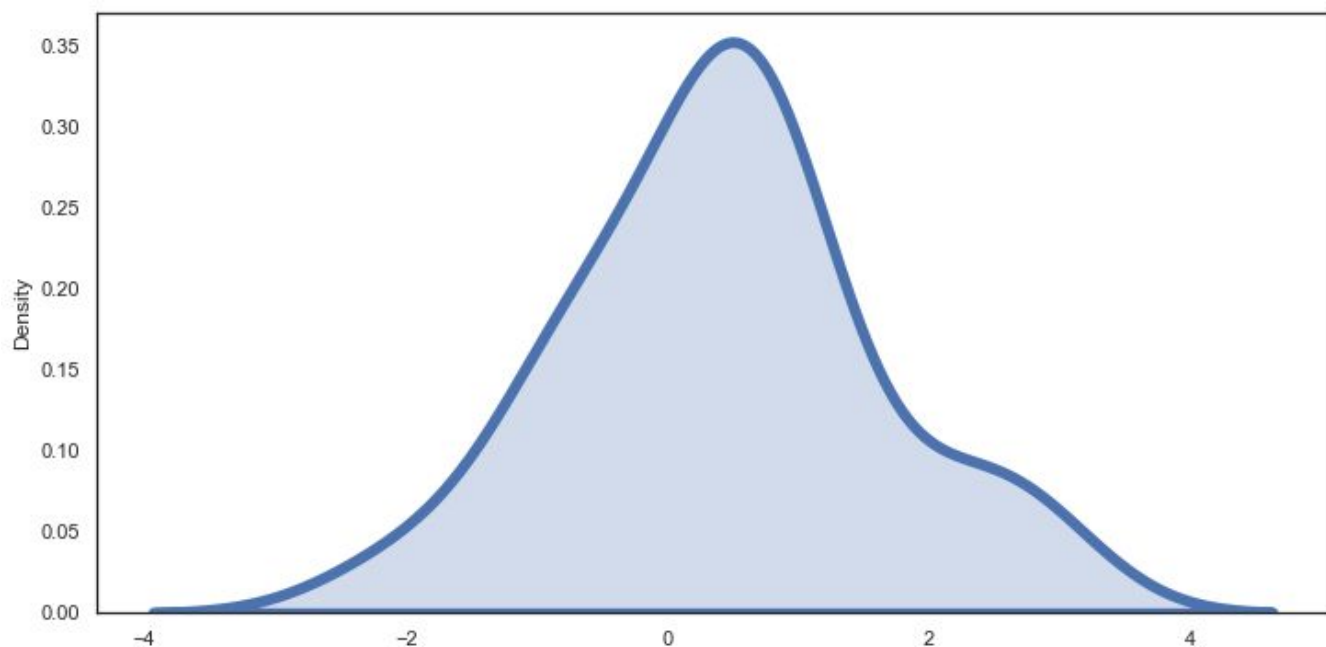
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You can change the kernel and bandwidth used which can make your KDE show more or less of the variance contained in the data.

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Categorical Plots

Statistical Estimation within Categories
Part One: Understanding the Plots

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The categorical plots display a statistical metrics **per** a category.

- For example mean value per category or a count of the number of rows **per** category.
- It is the visualization equivalent of a `groupby()` call.

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Two main types of plots for this are:

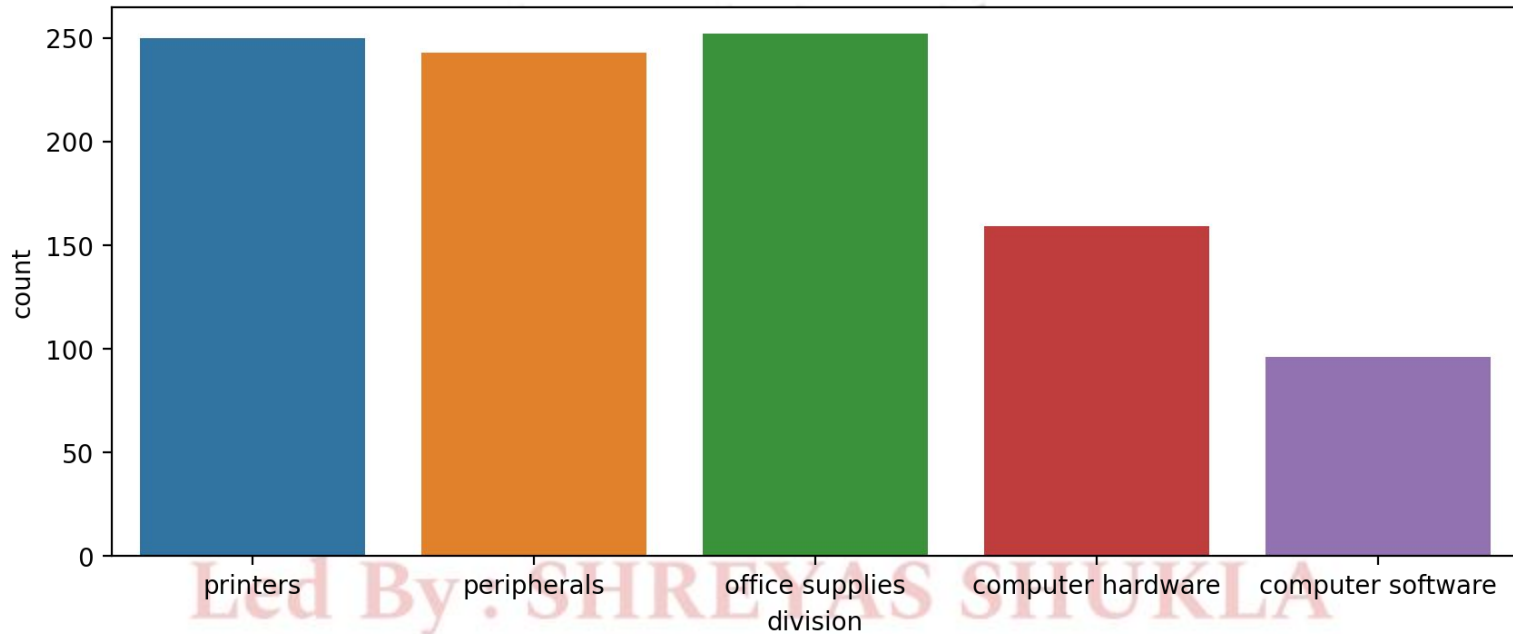
- `countplot()`
 - Counts number of rows per category.
- `barplot()`
 - General form of displaying any chosen metric per category.

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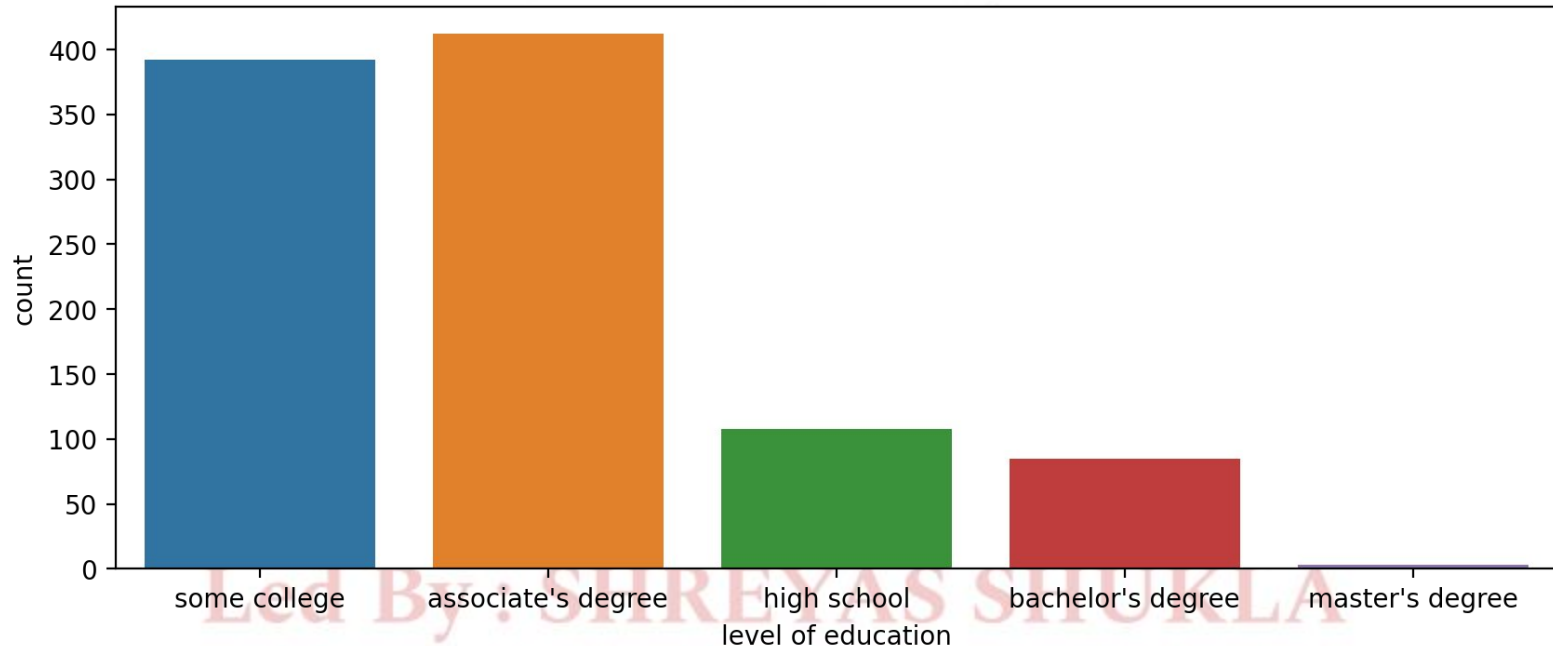
Countplot for corporate divisions



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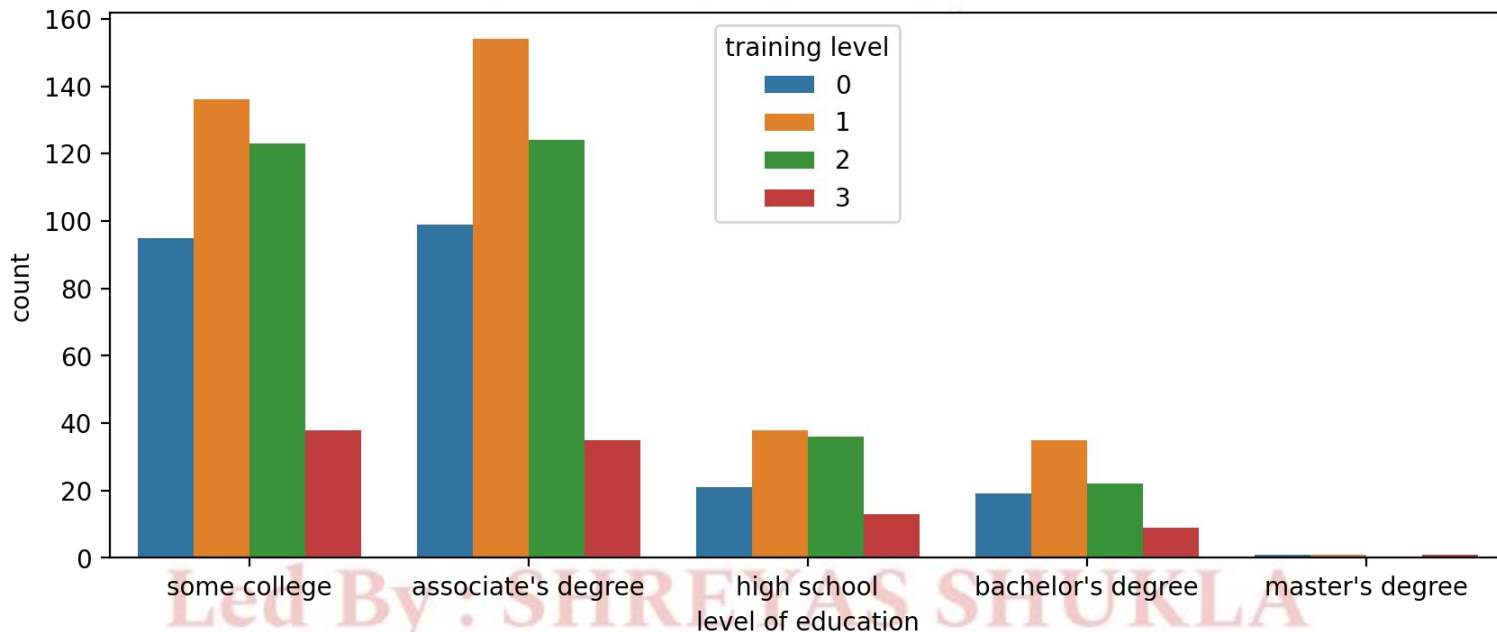
Countplot for education level



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Countplot with additional hue separation



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The barplot is the general form that allows you to choose any measure or estimator for the y axis.

We could plot the mean value and standard deviation per category instead.

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Caution!

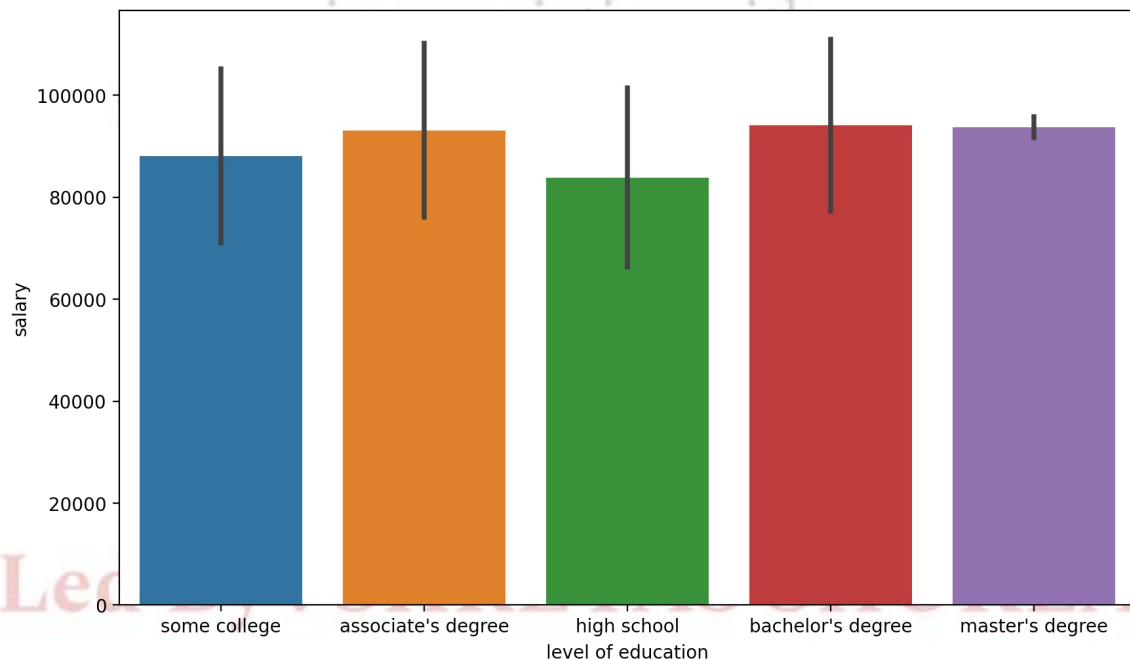
- Be very careful with these plots, since the bar is filled and continuous, a viewer may interpret continuity along the y-axis which may be incorrect!
- Always make sure to add additional labeling and explanation for these plots!

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Barplot showing mean and SD bar



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A simple table is probably better.

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if mean std

level of education

associate's degree	93156.41	17066.06
bachelor's degree	94133.76	17007.09
high school	83887.35	17674.44
master's degree	93718.00	2497.63
some college	88115.84	17076.28

Led

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Let's explore coding out these plots with seaborn!

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We've explored distribution plots for a single feature, but what if we want to compare distributions across categories?

For example, instead of the distribution of everyone's salary, we can compare the distributions of salaries **per** level of education.

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We will first separate out each category, then create the distribution visualization.

Let's explore what plot types we have available....

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Distribution within Categories

- Boxplot
- Violinplot
- Swarmplot

Let's explore understanding these plots on the previous salary dataset.

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The Boxplot displays the distribution of a continuous variable. It does this through the use of quartiles.

Quartiles separate out the data into 4 equal number of data points :

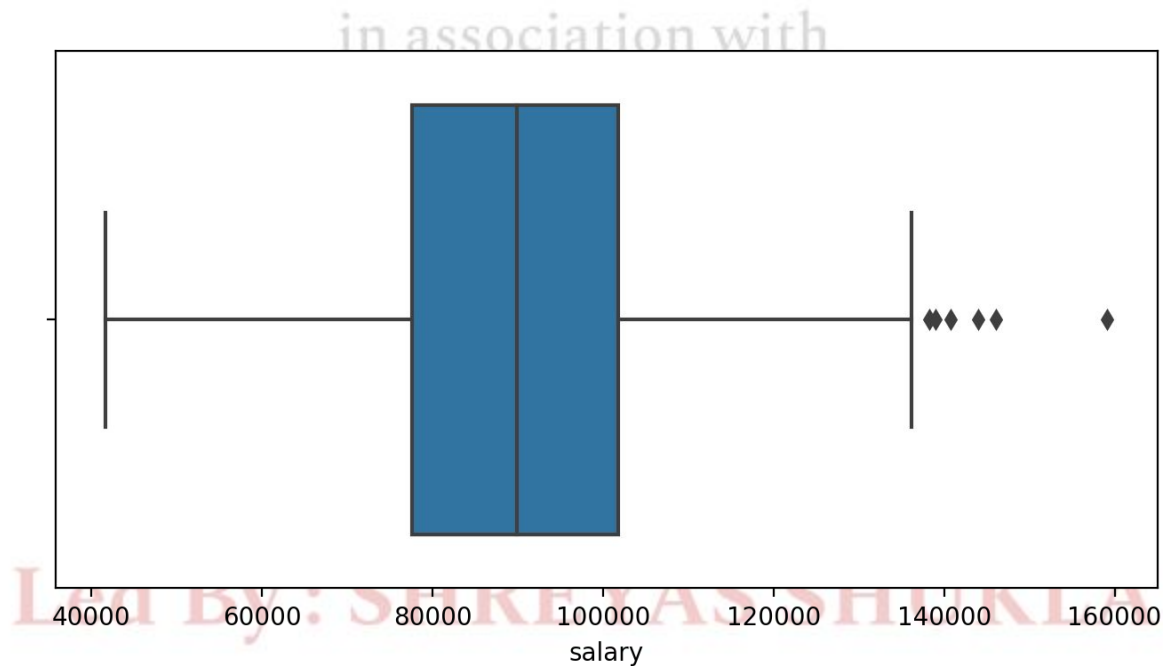
- 25% of data points are in bottom quartile.
- 50th percentile (Q2) is the median.

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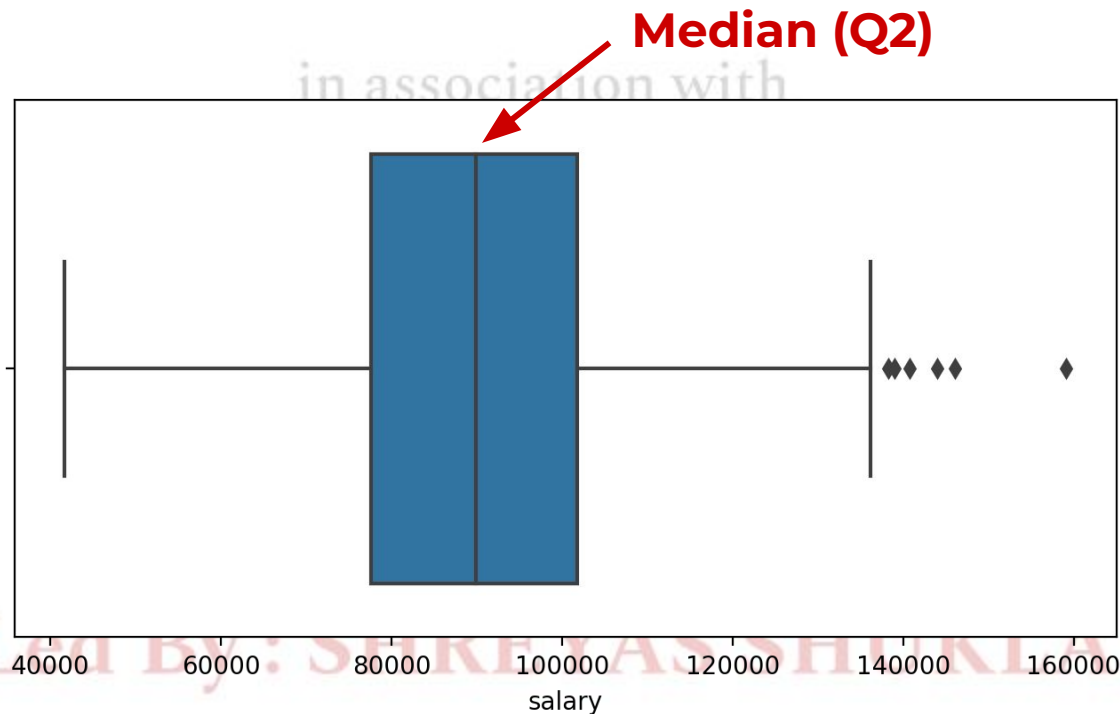
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Boxplot on single feature:

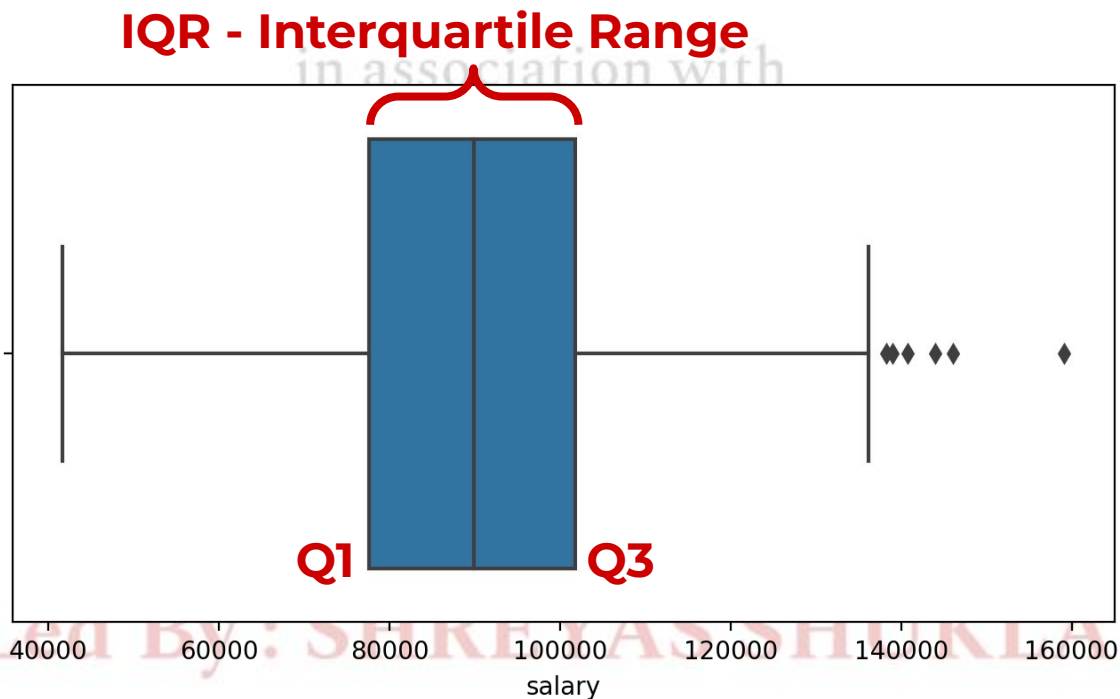


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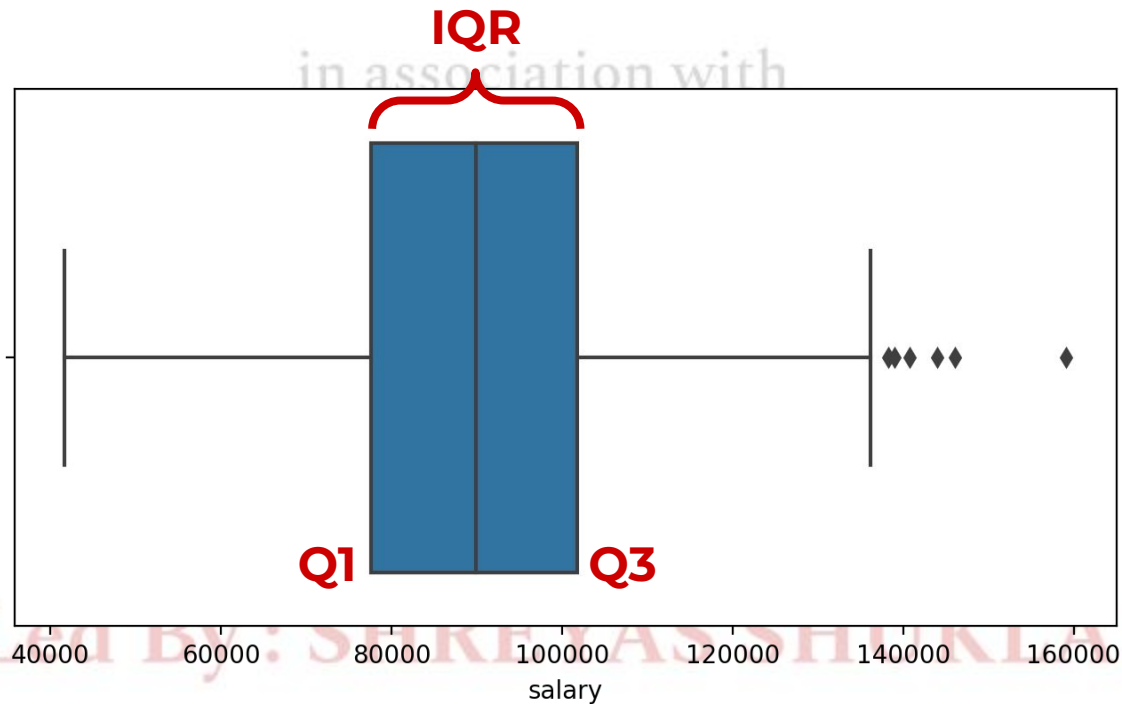
Median is 50th percentile. Median splits data in half



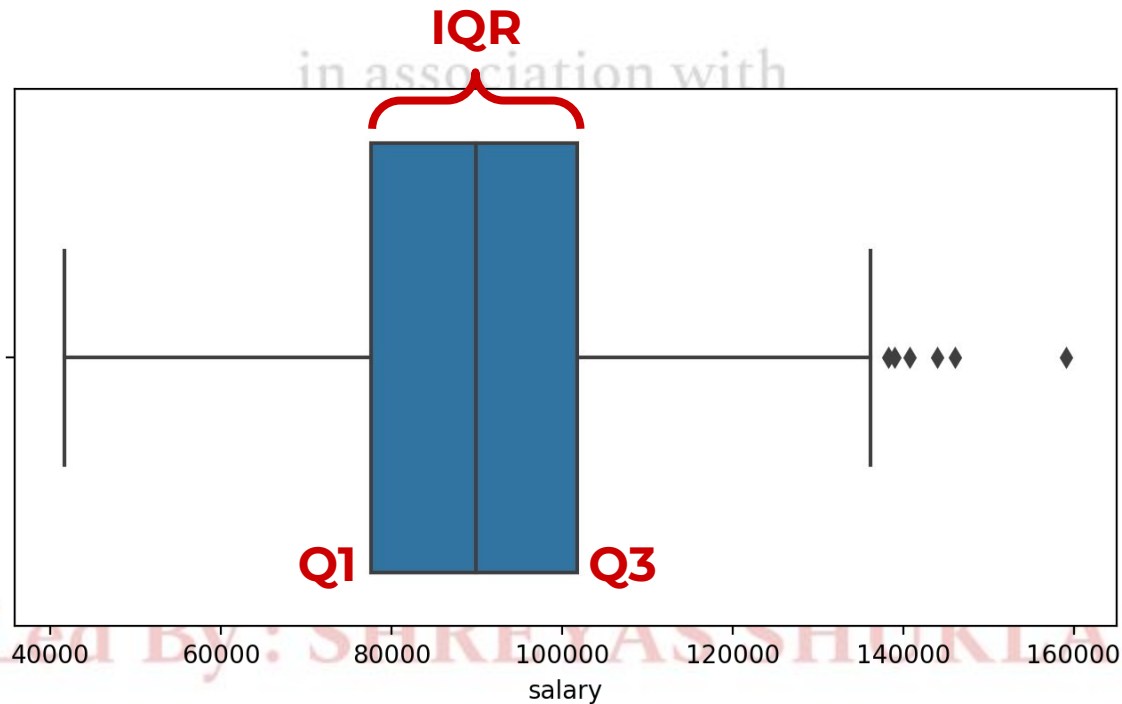
- IQR defines the box width
- 50% of all data points are inside the box



Q1 is the 25th percentile below which are 25% of data points



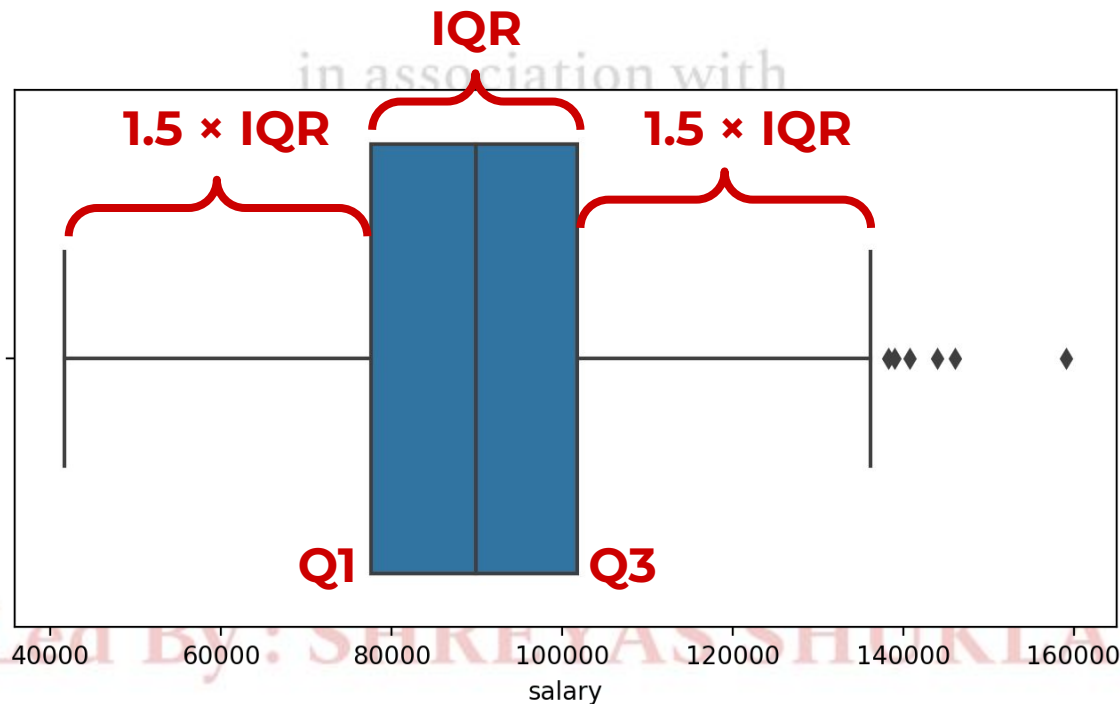
Q3 is the 75th percentile. 25% of all points are above Q3



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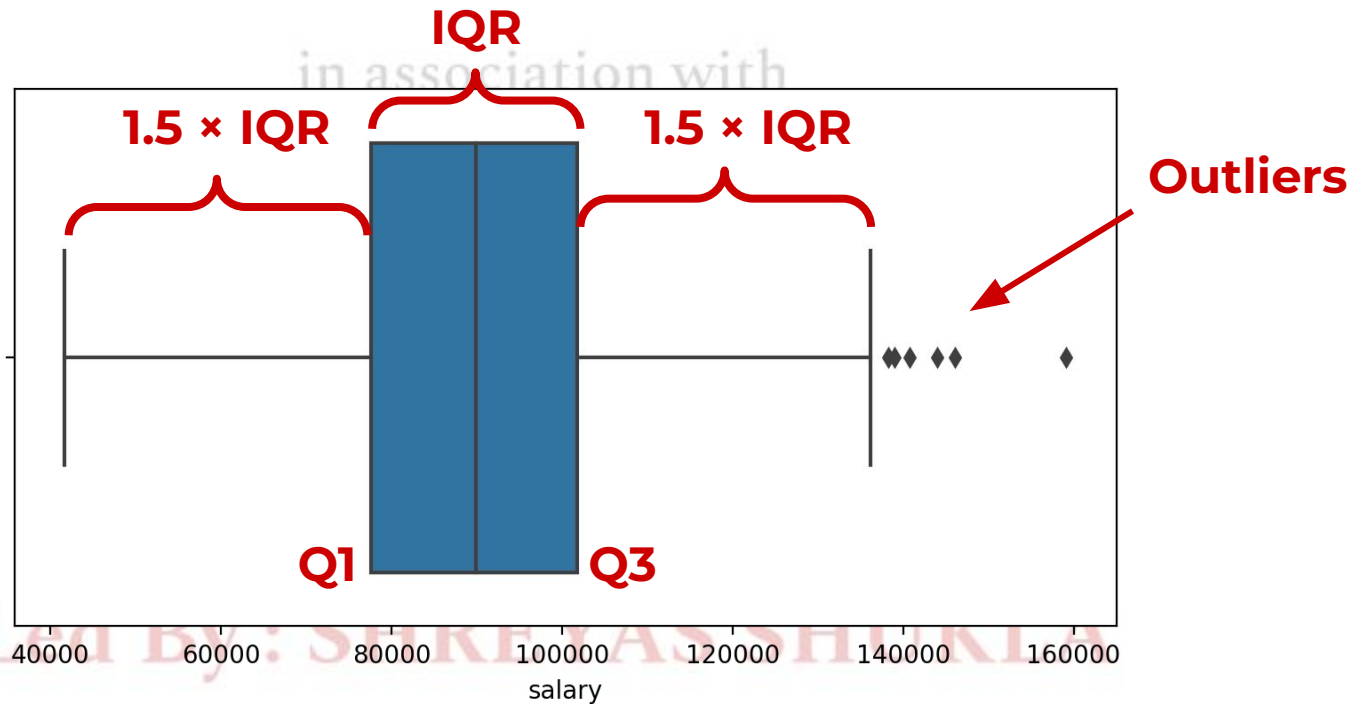
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The “whiskers” are defined by $1.5 \times \text{IQR}$



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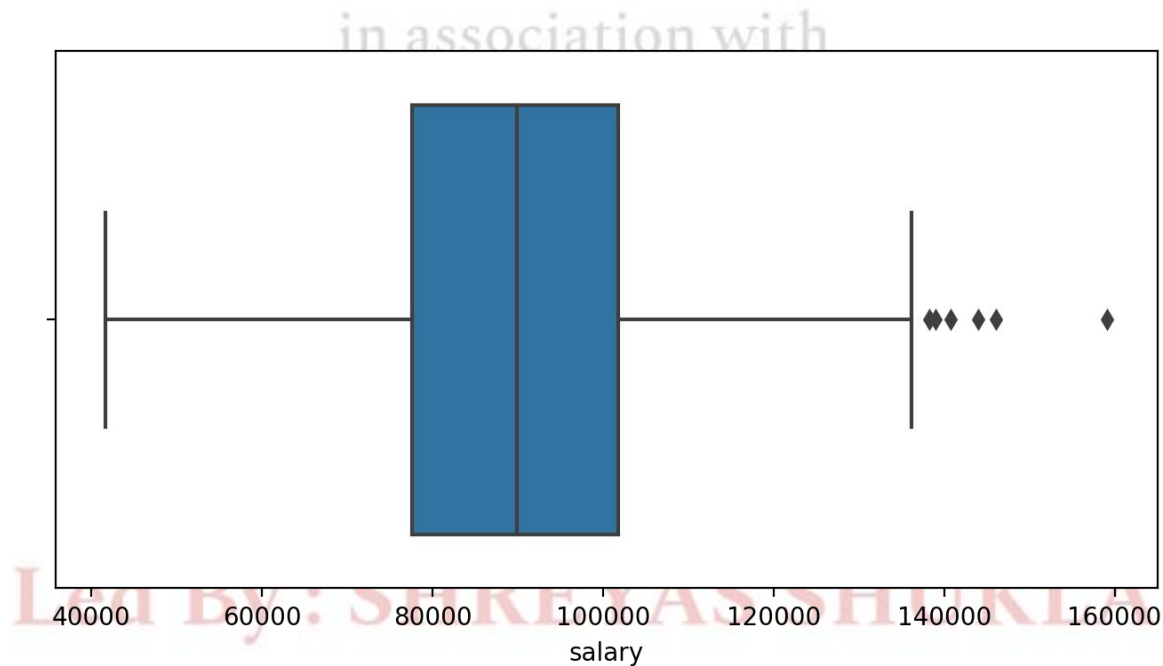
Outside of the whiskers are outliers



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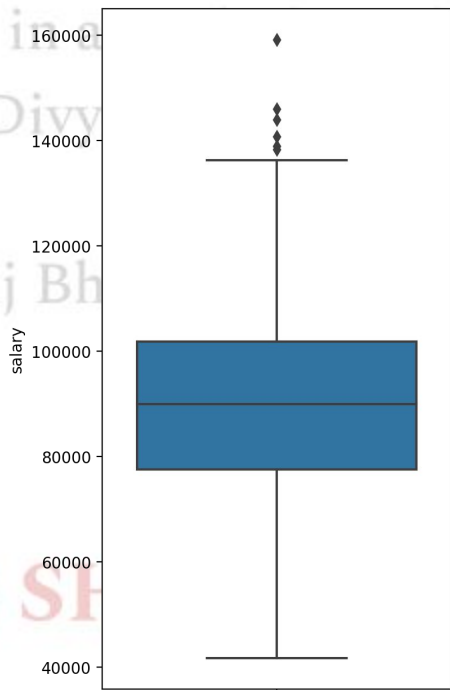
Boxplot gives statistical distribution information in a visual format:



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Boxplot can be oriented vertically or horizontally.



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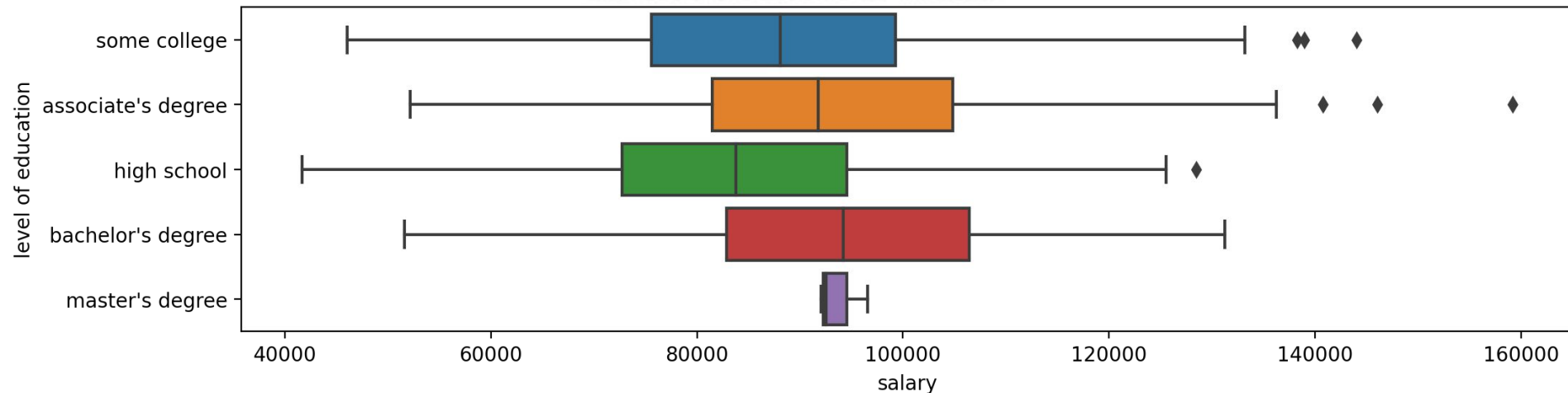
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We can create a box plot **per** category!

in association with

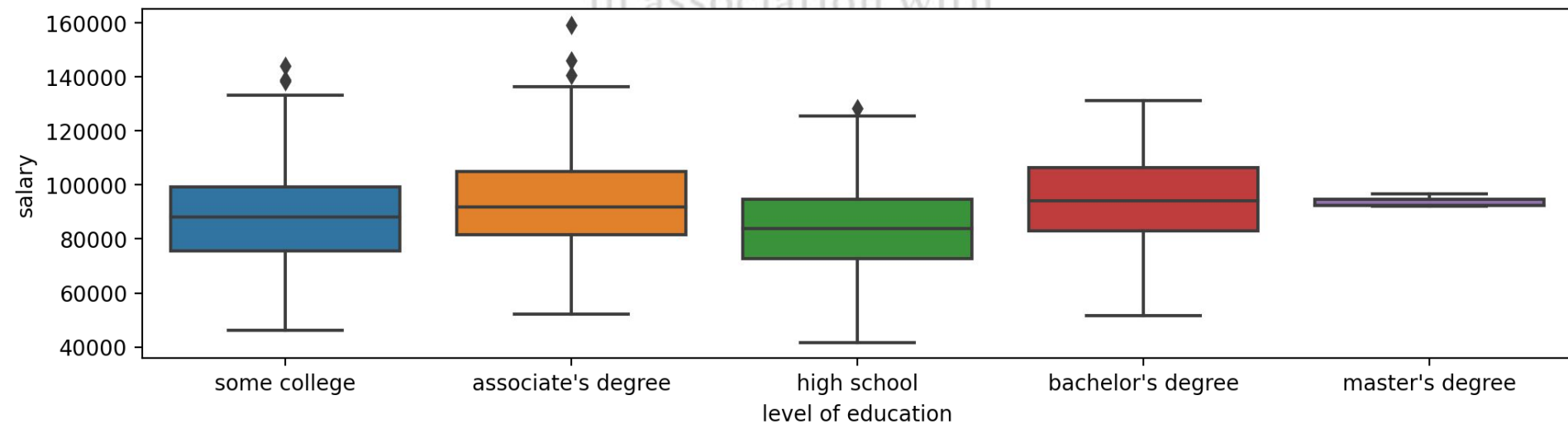


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The violin plot plays a similar role as the box plot. It displays the probability density across the data using a KDE.

We can imagine it as a mirrored KDE plot.

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The violin plot plays a similar role as the box plot. It displays the probability density across the data using a KDE.

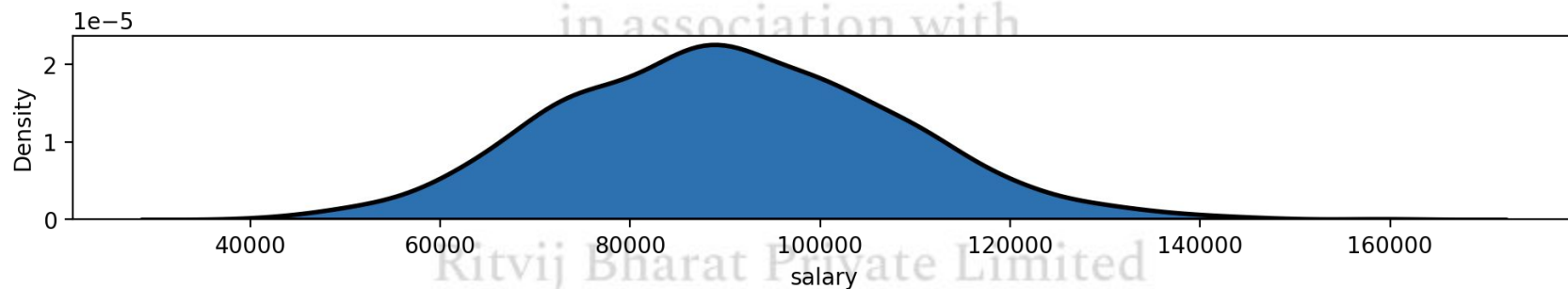
We can imagine it as a mirrored KDE plot.

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We take the KDE of a single feature:

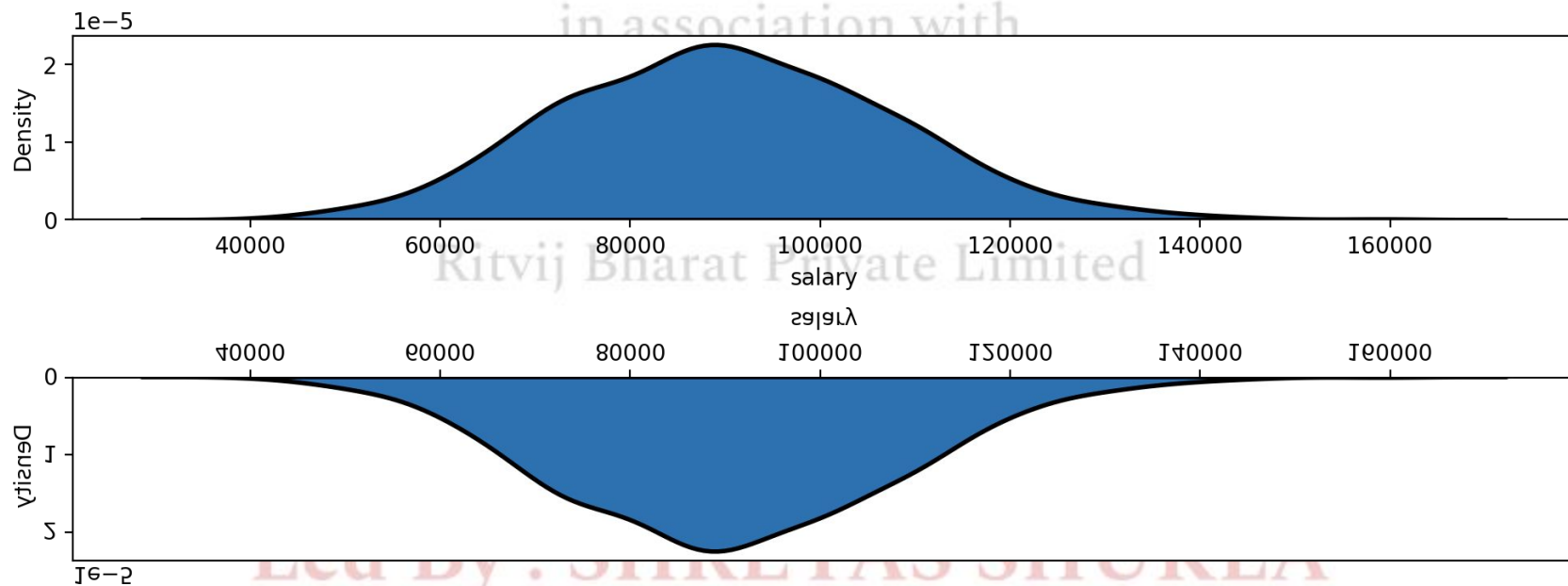


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We could then “mirror” it:



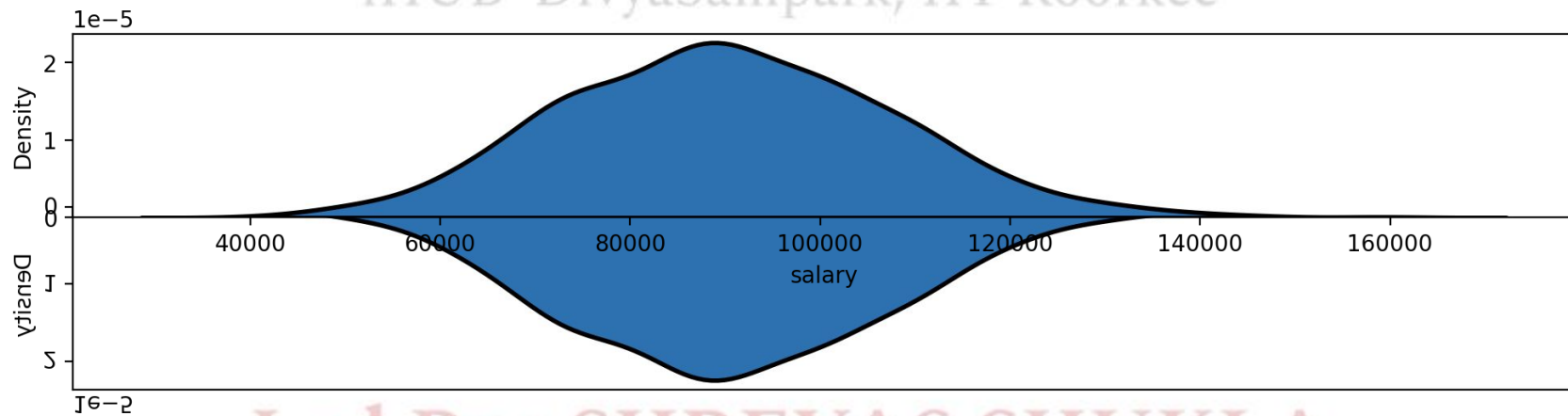
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Then combine it to get the violin plot:

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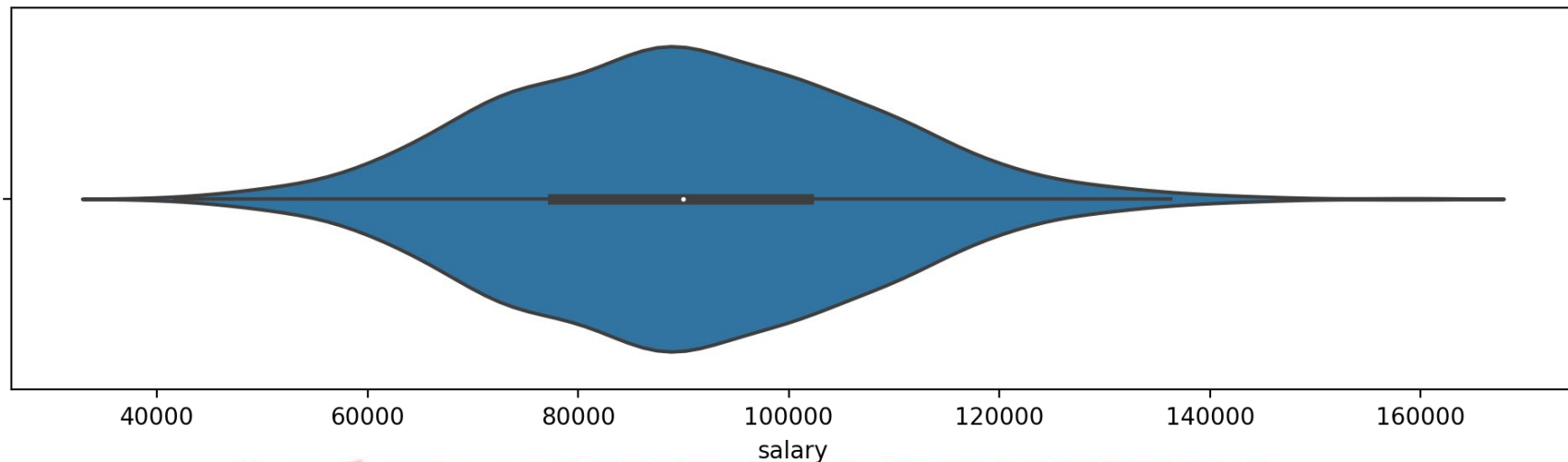
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in association with



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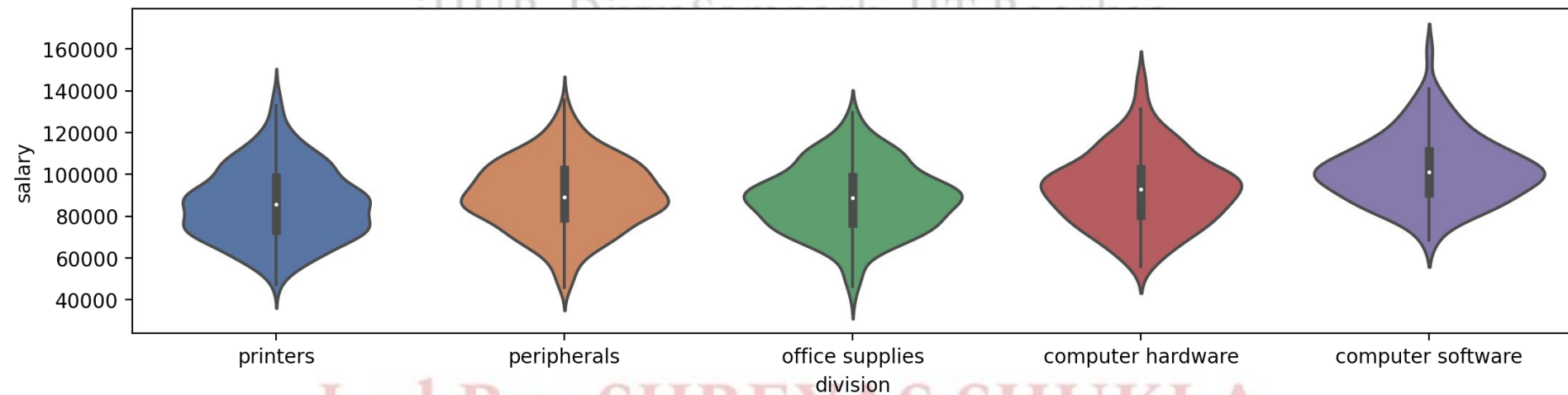
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The violin plots can then be created **per** category:

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'HUB.D' - C - 1 - HUB - 1



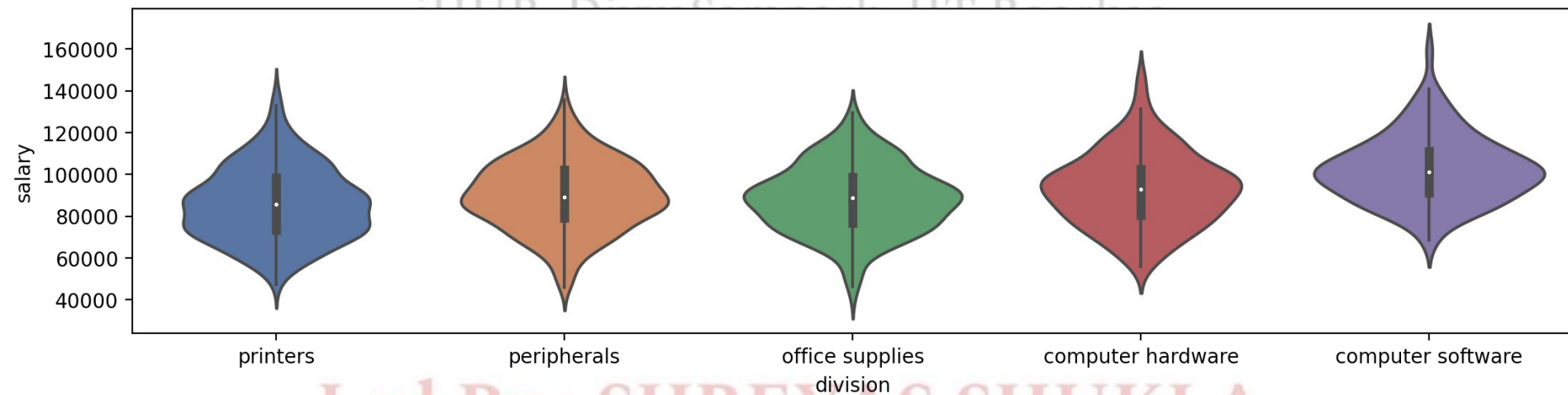
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A few more less common categorical distribution plot,
that is, swarmplot.

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Let's quickly explore this plot type...

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The swarmplot simply shows all the data points in the distribution.

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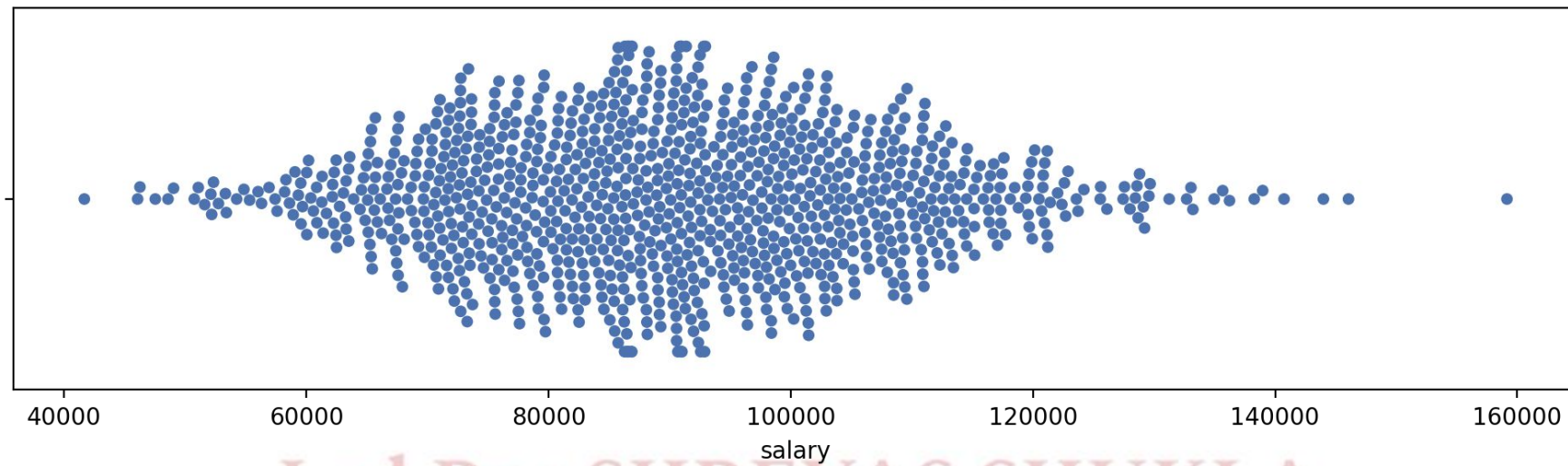
(For very large data sets, it won't show all the points, but will display the general distribution of them.)

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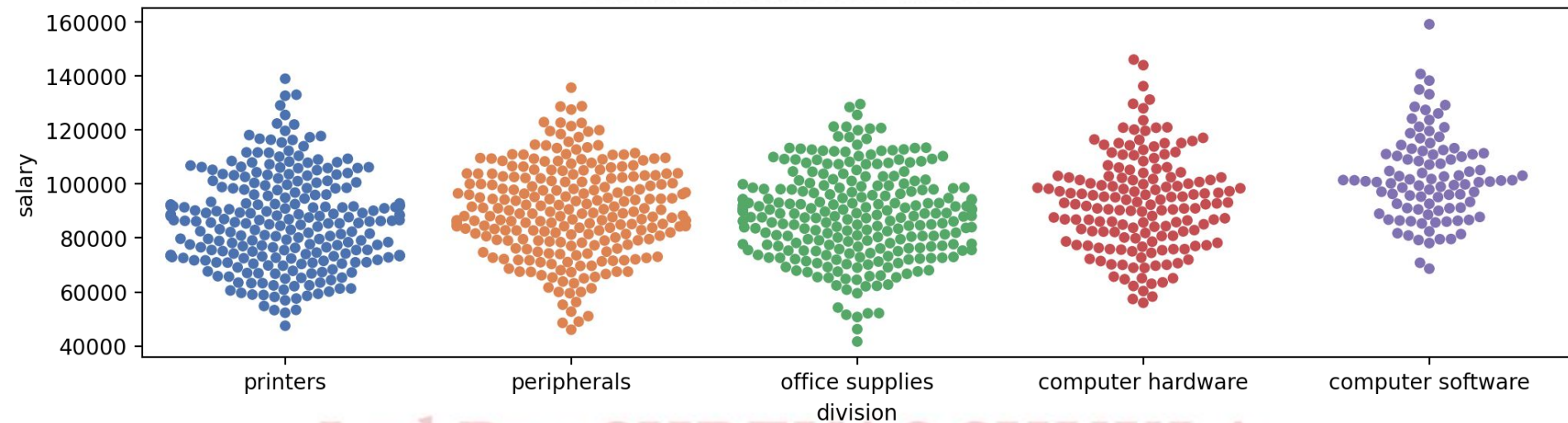


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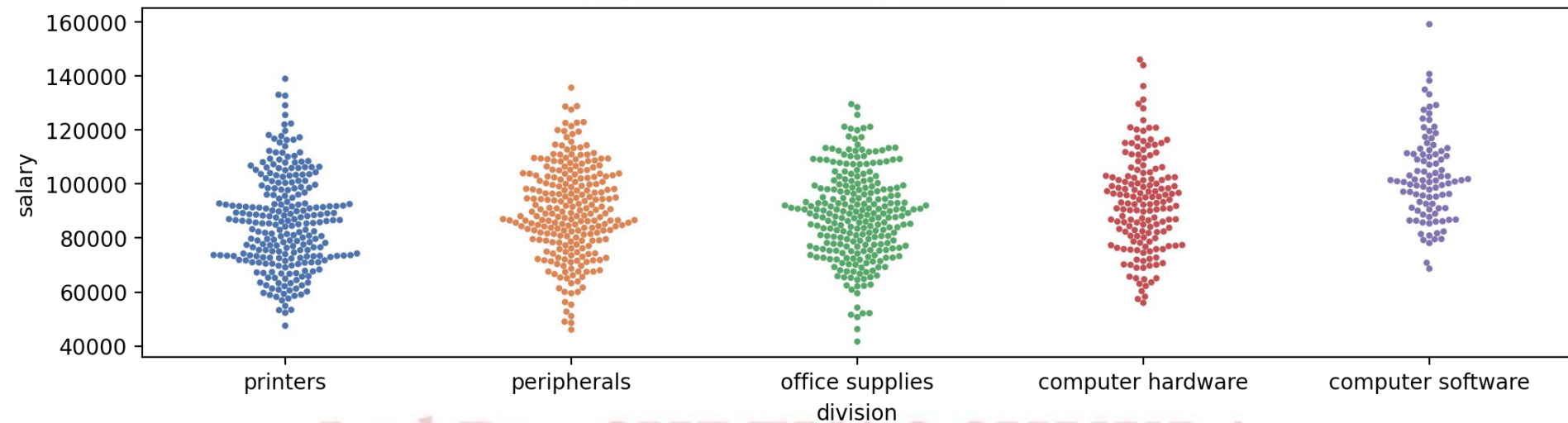


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That is it!!

Let's code !!

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